

Commutative Algebra notes

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Abstract

Notes taken while studying Commutative Algebra, mostly from Atiyah & MacDonald book [1] and Reid's book [2]. For the exercises, I follow the assignments listed at [3].

Usually while reading books and papers I take handwritten notes in a notebook, this document contains some of them re-written to *LaTeX*.

The proofs may slightly differ from the ones from the books, since I try to extend them for a deeper understanding.

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1 Ideals

1.1 Definitions

Definition (ideal). $I \subset R$ (R ring) such that $0 \in I$ and $\forall x \in I, r \in R, xr, rx \in I$.

ie. I absorbs products in R .

Definition (prime ideal). if $a, b \in R$ with $ab \in P$ and $P \neq R$ (P a prime ideal), implies $a \in P$ or $b \in P$.

Definition (principal ideal). generated by a single element, (a) .

(a) : principal ideal, the set of all multiples xa with $x \in R$.

Definition (maximal ideal). $\mathfrak{m} \subset A$ (A ring) with $\mathfrak{m} \neq A$ and there is no ideal I strictly between $\mathfrak{m}$ and A . ie. if $\mathfrak{m}$ maximal and $\mathfrak{m} \subseteq I \subseteq A$, either $\mathfrak{m} = I$ or $I = A$.

Definition (unit). $x \in A$ such that $xy = 1$ for some $y \in A$. ie. element *which divides 1*.

Corollary 1.8. $A = A^\times \sqcup \bigcup \mathfrak{m}$ (where $\sqcup$ denotes “disjoint union”), ie. $f \in A$ is either a unit or it is contained in a maximal ideal, not both.

Definition (zerodivisor). $x \in A$ such that $\exists 0 \neq y \in A$ such that $xy = 0 \in A$. ie. x *divides 0*.

If a ring does not have zerodivisors is an integral domain.

Definition (prime spectrum - $\text{Spec}(A)$). set of prime ideals of A . ie.

$$\text{Spec}(A) = \{P \mid P \subset A \text{ is a prime ideal}\}$$

Definition (integral domain). Ring in which the product of any two nonzero elements is nonzero.

ie. no zerodivisors.

ie. $\forall 0 \neq a, 0 \neq b \in A, ab \neq 0 \in A$.

Every field is an integral domain, not the converse.

Definition (principal ideal domain - PID). integral domain in which every ideal is principal. ie. $\forall I \subset R, \exists a \in I$ such that $I = (a) = \{ra \mid r \in R\}$.

Definition (nilpotent). $a \in A$ such that $a^n = 0$ for some $n > 0$.

Definition (nilrad A). set of all nilpotent elements of A ; is an ideal of A .
if $\text{nilrad} A = 0 \implies A$ has no nonzero nilpotents.

$$\text{nilrad} A = \bigcap_{P \in \text{Spec}(A)} P$$

Definition (idempotent). $e \in A$ such that $e^2 = e$.

Definition (radical of an ideal).

$$\text{rad} I = \{f \in A \mid f^n \in I \text{ for some } n\}$$

$\text{rad} I$ is an ideal.

$$\text{nilrad} A = \text{rad} 0$$

$$\text{rad} I = \bigcap_{\substack{P \in \text{Spec}(A) \\ P \supset I}} P$$

Definition 1.13 (local ring). A ring is *local* if it has a unique maximal ideal.

Notation: local ring A , its maximal ideal $\mathfrak{m}$, residue field $K = A/\mathfrak{m}$:

$$A \supset \mathfrak{m} \text{ or } (A, \mathfrak{m}) \text{ or } (A, \mathfrak{m}, K)$$

By Corollary 1.8, A is local

$\iff A$ has only one maximal ideal.

$\iff$ all the nonunits of A form an ideal.

Proposition ($R/\mathfrak{m} = \text{Field}$). Let R be a commutative ring with identity, and let $m \subsetneq R$ be a maximal ideal. Then the quotient ring R/m is a field.

Proof. Let $x + m$ be a nonzero element of R/m . Since

$$x + m \neq m$$

we have $x \notin m$.

Consider the ideal $m + (x)$, where (x) is the principal ideal generated by x . Since $x \notin m$, this ideal strictly contains m :

$$m \subsetneq m + (x)$$

Because m is maximal, there are no proper ideals strictly between m and R . Therefore, $m + (x) = R$

In particular, $1 \in m + (x)$. Hence, there exist $a \in m$ and $y \in R$ such that $a + yx = 1$.

Passing to the quotient modulo m , we obtain

$$(y + m)(x + m) = yx + m = 1 + m,$$

because $a \in m$ implies $a + m = m$.

Thus, $y + m$ is the multiplicative inverse of $x + m$. Therefore, every nonzero element of R/m is invertible, and hence R/m is a field. $\square$

1.2 $\mathbb{Z}$ and $K[X]$, two Principal Ideal Domains

Lemma . $\mathbb{Z}$ is a PID.

Proof. Let I a nonzero ideal of $\mathbb{Z}$.

Since $I \neq \{0\}$, there is at least one nonzero integer in I . Choose the smallest element of I , namely d .

Observe that $(d) \subseteq I$, since $d \in I$. Then, every multiple $nd \in I$, since I is an ideal.

Take $a \in I$. By the Euclidean division algorithm in $\mathbb{Z}$, $a = qd + r$, with $q, r \in \mathbb{Z}$ and $0 \leq r < d$.

Then $r = a - qd \in I$, but d was chosen to be the smallest positive element of I , so the only possibility is $r = 0$.

Hence, $a = qd$, so $a \in (d)$, giving $I \subseteq (d)$.

Since we had $(d) \subseteq I$ and now we got $I \subseteq (d)$, we have $I = (d)$, so every ideal of $\mathbb{Z}$ is principal. Thus $\mathbb{Z}$ is a Principal Ideal Domain(PID). $\square$

Lemma . $K[X]$ is a PID.

Proof. This proof follows very similarly to the previous proof.

Let K be a field, $K[X]$ a polynomial ring.

Take $\{0\} \neq I \subseteq K[X]$.

Since $I \neq \{0\}$, there is at least one non-zero polynomial in I .

Let $p(X) \in I$ be of minimal degree among nonzero elements of I .

Observe that $(p(X)) \subseteq I$, because $p(X) \in I$ and I is an ideal.

Let $f(X) \in I$. By Euclidean division algorithm in $K[X]$, $\exists q, r \in K[X]$ such that $f(X) = q(X) \cdot p(X) + r(X)$ with either $r(X) = 0$ or $\deg(r) < \deg(p)$.

Since $f, p \in I$, then $r(X) = f(X) - q(X) \cdot p(X) \in I$

If $r(X) \neq 0$, then $\deg(r) < \deg(p)$, which contradicts the minimality of $\deg(p)$ in I .

Therefore, $r(X) = 0$, thus $f(X) = q(X) \cdot p(X)$, hence $f(X) \in (p(X))$. Henceforth, $I \subseteq (p(X))$.

Then, since $(p(X)) \subseteq I$ and $I \subseteq (p(X))$, we have that $I = (p(X))$.

So every ideal of $K[X]$ is principal; thus $K[X]$ is a PID. $\square$

1.3 Zorn's lemma and Jacobson radicals

Let Σ be a partially ordered set. Given subset $S \subset \Sigma$, an *upper bound* of S is an element $u \in \Sigma$ such that $s < u \forall s \in S$.

A *maximal element* of Σ , is $m \in \Sigma$ such that $m < s$ does not hold for any $s \in \Sigma$.

A subset $S \subset \Sigma$ is *totally ordered* if for every pair $s_1, s_2 \in S$, either $s_1 \leq s_2$ or $s_2 \leq s_1$.

Lemma R.1.7 (Zorn's lemma). Suppose Σ a nonempty partially ordered set (ie. we are given a relation $x \leq y$ on Σ), and that any totally ordered subset $S \subset \Sigma$ has an upper bound in Σ .

Then Σ has a maximal element.

Theorem AM.1.3. Every ring $A \neq 0$ has at least one maximal ideal.

Proof. By Zorn's lemma R.1.7. □

Corollary AM.1.4. if $I \neq (1)$ an ideal of A , $\exists$ a maximal ideal of A containing I .

Corollary AM.1.5. Every non-unit of A is contained in a maximal ideal.

Definition (Jacobson radical). The *Jacobson radical* of a ring A is the intersection of all the maximal ideals of A .

Denoted $Jac(A)$.

$Jac(A)$ is an ideal of A .

Proposition AM.1.9. $x \in Jac(A)$ iff $(1 - xy)$ is a unit in A , $\forall y \in A$.

Proof. Suppose $1 - xy$ not a unit.

By AM.1.5, $1 - xy \in \mathfrak{m}$ for $\mathfrak{m}$ some maximal ideal.

But $x \in Jac(A) \subseteq \mathfrak{m}$, since $Jac(A)$ is the intersection of all maximal ideals of A .

Hence $xy \in \mathfrak{m}$, and therefore $1 \in \mathfrak{m}$, which is absurd, thus $1 - xy$ is a unit.

Conversely:

Suppose $x \notin \mathfrak{m}$ for some maximal ideal $\mathfrak{m}$.

Then $\mathfrak{m}$ and x generate the unit ideal (1) , so that we have $u + xy = 1$ for some $u \in \mathfrak{m}$ and some $y \in A$.

Hence $1 - xy \in \mathfrak{m}$, and is therefore not a unit. □

2 Modules

2.1 Modules concepts

Let A be a ring. An A -module is an Abelian group M with a multiplication map

$$\begin{aligned} A \times M &\longrightarrow M \\ (f, m) &\longmapsto fm \end{aligned}$$

satisfying $\forall f, g \in A, m, n \in M$.

- i. $f(m \pm n) = fm \pm fn$
- ii. $(f \pm g)m = fm \pm gm$
- iii. $(fg)m = f(gm)$
- iv. $1_A m = m$

Let $\psi : M \rightarrow M$ an A -linear endomorphism of M .
 $A[\psi] \subset \text{End} M$ is the subring generated by A and the action of ψ .

- since ψ is A -linear, $A[\psi]$ is a commutative ring.
- M is a module over $A[\psi]$, so ψ becomes multiplication by a ring element.

Remark . Vector spaces are defined over fields, a Module is the generalization of a vector space where the scalars come from a ring instead of a field.

2.2 Cayley-Hamilton theorem, Nakayama lemma, and corollaries

Proposition AM.2.4. (Cayley-Hamilton Theorem) Let M a fingen A -module. Let $\mathfrak{a}$ an ideal of A , let ψ an A -module endomorphism of M such that $\psi(M) \subseteq \mathfrak{a}M$.

Then ψ satisfies

$$\psi^n + a_1 \psi^{n-1} + \dots + a_{n-1} \psi + a_n = 0$$

with $a_i \in \mathfrak{a}$.

Proof. Since M fingen, let $\{x_1, \dots, x_n\}$ be generators of M .

By hypothesis, $\psi(M) \subseteq \mathfrak{a}M$; so for any generator x_i , it's image $\psi(x_i) \in \mathfrak{a}M$.

Any element in $\mathfrak{a}M$ is a linear combination of the generators with coefficients in the ideal $\mathfrak{a}$, thus

$$\psi(x_i) = \sum_{j=1}^n a_{ij} x_j$$

with $a_{ij} \in \mathfrak{a}$.

Thus, for a module with n generators, we have n different $\psi(x_i)$ equations:

$$\left. \begin{array}{l} \psi(x_1) = a_{1,1}x_1 + a_{1,2}x_2 + \dots + a_{1,n}x_n \\ \psi(x_2) = a_{2,1}x_1 + a_{2,2}x_2 + \dots + a_{2,n}x_n \\ \dots \\ \psi(x_n) = a_{n,1}x_1 + a_{n,2}x_2 + \dots + a_{n,n}x_n \end{array} \right\} \begin{array}{l} n \text{ elements } \psi(x_i) \in \mathfrak{a}M \text{ which} \\ \text{are linear combinations of the} \\ n \text{ generators of } M \end{array}$$

Next step: rearrange in order to use matrix algebra.

Observe that each row equals 0, and rearranging the elements at each row we get

$$\left. \begin{aligned} \psi(x_1) - (a_{1,1}x_1 + a_{1,2}x_2 + \dots + a_{1,n}x_n) &= 0 \\ \psi(x_2) - (a_{2,1}x_1 + a_{2,2}x_2 + \dots + a_{2,n}x_n) &= 0 \\ \dots \\ \psi(x_n) - (a_{n,1}x_1 + a_{n,2}x_2 + \dots + a_{n,n}x_n) &= 0 \end{aligned} \right\}$$

Then, group the x_i terms together; as example, take the row $i = 1$:

$$(\psi - a_{1,1})x_1 - a_{1,2}x_2 - \dots - a_{1,n}x_n = 0$$

$$\left. \begin{aligned} (\psi - a_{1,1})x_1 - a_{1,2}x_2 - \dots - a_{1,n}x_n &= 0 \\ -a_{2,1}x_1 + (\psi - a_{2,2})x_2 - \dots - a_{2,n}x_n &= 0 \\ \dots \\ -a_{n,1}x_1 - a_{n,2}x_2 - \dots + (\psi - a_{n,n})x_n &= 0 \end{aligned} \right\}$$

So, $\forall i \in [n]$, as a matrix:

$$\begin{pmatrix} \psi - a_{1,1} & -a_{1,2} & \dots & -a_{1,n} \\ -a_{2,1} & \psi - a_{2,2} & \dots & -a_{2,n} \\ \vdots & & & \\ -a_{n,1} & -a_{n,2} & \dots & \psi - a_{n,n} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

Denote the previous matrix by Φ . Let m denote the vector $(x_1, x_2, \dots, x_n)^T$ (ie. the vector of generators of the A -module M).

Then we can write the previous equality as

$$\Phi \cdot m = 0 \tag{1}$$

We know that

$$\text{adj}(\Phi)\Phi = \det(\Phi)I \tag{2}$$

(aka. *fundamental identity for the adjugate matrix*).

So if at (1) we multiply both sides by $\text{adj}(\Phi)$,

$$\begin{aligned} \text{adj}(\Phi) \cdot \Phi \cdot m &= 0 \\ (\text{recall from (2): } \text{adj}(\Phi)\Phi &= \det(\Phi) \cdot I) \\ &= \det(\Phi) \cdot I \cdot m = 0 \end{aligned}$$

Thus,

$$\begin{aligned} &\det(\Phi) \cdot I \cdot m = 0 : \\ &\begin{pmatrix} \det(\Phi) & 0 & \dots & 0 \\ 0 & \det(\Phi) & \dots & 0 \\ \vdots & & & \\ 0 & 0 & \dots & \det(\Phi) \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \end{aligned}$$

$\Rightarrow$

$$\det(\Phi) \cdot x_i = 0 \quad \forall i \in [n] \quad (3)$$

ie. $\det(\Phi)$ is an *annihilator* of the generators x_i of M , thus is an annihilator of the entire module M .

So, we're interested into calculating the $\det(\Phi)$.

By the Leibniz formula,

$$\det(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n a_{i,\sigma(i)}$$

thus,

$$\det(\Phi) = \underbrace{(\psi - a_{11})(\psi - a_{22}) \dots (\psi - a_{nn})}_{\text{diagonal of } \Phi, \text{ leading term of the determinant}} - \dots$$

The *determinant trick* is that the terms that go after the "leading term of the determinant", will belong to $\mathfrak{a}$ and their combinations with ψ will not be bigger than ψ^n . Furthermore, when expanding it

- highest power is $1 \cdot \psi^n$
- coefficient of ψ^{n-1} is $-(\underbrace{a_{11} + a_{22} + \dots + a_{nn}}_{a_1})$,

where, since each $a_{ii} \in \mathfrak{a}$, $a_1 \in \mathfrak{a}$

- the rest of coefficients of ψ^k are also elements in $\mathfrak{a}$

Therefore we have

$$\det(\Phi) = \psi^n + a_1 \psi^{n-1} + a_2 \psi^{n-2} + \dots + a_{n-1} \psi + a_n$$

with $a_i \in \mathfrak{a}$.

Now, notice that we had $\det(\Phi) \cdot x_i = 0 \quad \forall i \in [n]$.

The matrix Φ is the *characteristic matrix*, $xI - A$, viewed as an operator.

Then,

$$\det(\Phi) = \det(xI - A) = p(x)$$

where $p(x)$ is the *characteristic polynomial*.

If a linear transformation turns every basis vector (x_i) into zero, then that transformation is the zero transformation. So in our case, $\det(\Phi)$ is the zero transformation, thus $\det(\Phi) = 0$. Therefore,

$$\psi^n + a_1 \psi^{n-1} + a_2 \psi^{n-2} + \dots + a_{n-1} \psi + a_n = 0$$

□

Corollary AM.2.5. Let M a fingen A -module, let $\mathfrak{a}$ an ideal of A such that $\mathfrak{a}M = M$.

Then, $\exists x \equiv 1 \pmod{\mathfrak{a}}$ such that $xM = 0$.

Proof. take $\psi = \text{identity}$. Then in Cayley-Hamilton (AM.2.4):

$$\begin{aligned}\psi^n + a_1\psi^{n-1} + a_2\psi^{n-2} + \dots + a_{n-1}\psi + a_n &= 0 \\ \implies id_M + a_1id_M + a_2id_M + \dots + a_{n-1}id_M + a_n &= 0 \\ \implies (1 + a_1 + \dots + a_n)id_M &= 0\end{aligned}$$

apply it to $m \in M$, where since $id_M(m) = m$ (by definition of the identity), we then have

$$(1 + a_1 + \dots + a_n) \cdot m = 0$$

with $a_i \in \mathfrak{a}$.

part i. $xM = 0$:

Thus the scalar $x = (1 + a_1 + \dots + a_n)$ annihilates every $m \in M$, ie. the entire module M .

part ii. $x \equiv 1 \pmod{\mathfrak{a}}$:

$$x \equiv 1 \pmod{\mathfrak{a}} \iff (x - 1) \in \mathfrak{a}$$

then from $x = (1 + \underbrace{a_1 + \dots + a_n}_b) \in \mathfrak{a}$, set $b = a_1 + \dots + a_n$,

so that $x = (1 + b) \in \mathfrak{a}$.

Then $x - 1 = (1 + b) - 1 = b \in \mathfrak{a}$

so $x - 1 \in \mathfrak{a}$, thus $x \equiv 1 \pmod{\mathfrak{a}}$ as stated.

□

Proposition AM.2.6 (Nakayama's lemma). Let M a fingen A -module, let $\mathfrak{a}$ an ideal of A such that $\mathfrak{a} \subseteq \text{Jac}(A)$.

Then $\mathfrak{a}M = M$ implies $M = 0$.

Proof. By AM.2.5: since $\mathfrak{a}M = M$, we have $xM = 0$ for some $x \equiv 1 \pmod{\text{Jac}(A)}$. (notice that at AM.2.5 is $\pmod{\mathfrak{a}}$ but here we use $\pmod{\text{Jac}(A)}$), since we have $\mathfrak{a} \subseteq \text{Jac}(A)$).

(recall AM.1.9: $x \in \text{Jac}(A)$ iff $(1 - xy)$ is a unit in A , $\forall y \in A$).

By AM.1.9, x is a unit in A (thus $x^{-1} \cdot x = 1$).

$$\begin{aligned}\text{Hence } M &= x^{-1} \cdot \underbrace{x \cdot M}_{=0 \text{ (by AM.2.5)}} = 0.\end{aligned}$$

Thus, if $\mathfrak{a}M = M$ then $M = 0$.

□

Corollary AM.2.7. Let M a fingen A -module, let $N \subseteq M$ a submodule of M , let $\mathfrak{a} \subseteq \text{Jac}(A)$ an ideal.

Then $M = \mathfrak{a}M + N \xrightarrow{\text{implies}} M = N$.

Proof. The idea is to apply Nakayama (AM.2.6) to M/N .
 Since M fingen $\implies M/N$ is fingen and an A -module.
 Since $\mathfrak{a} \subseteq \text{Jac}(A) \implies$ Nakayama applies to M/N too.
 By definition,

$$\mathfrak{a}M = \left\{ \sum a_i \cdot m_i \mid a_i \in \mathfrak{a}, m_i \in M \right\}$$

where m_i are the generators of M .

Then, for M/N ,

$$\mathfrak{a}\left(\frac{M}{N}\right) = \left\{ \sum a_i \cdot (m_i + N) \mid a_i \in \mathfrak{a}, m_i \in M \right\}$$

observe that $a_i(m_i + N) = a_i m_i + N$, thus

$$\sum_i a_i \cdot (m_i + N) = \underbrace{\left(\sum_i a_i \cdot m_i \right)}_{\in \mathfrak{a}M} + N \in \mathfrak{a}M + N$$

Hence,

$$\mathfrak{a}\left(\frac{M}{N}\right) = \{x + N \mid x \in \mathfrak{a}M\} = \mathfrak{a}M + N \quad (4)$$

By definition, if we take $\frac{\mathfrak{a}M + N}{N}$, then

$$\frac{\mathfrak{a}M + N}{N} = \{y + N \mid y \in \mathfrak{a}M + N\} = \mathfrak{a}M + N$$

thus every $y \in \mathfrak{a}M + N$ can be written as

$$y = x + n, \text{ with } x \in \mathfrak{a}M, n \in N$$

which comes from (4).

Thus, $y + N = (x + n) + N = x + N$, since $n \in N$ is zero in the quotient.

Hence, every element of $\frac{\mathfrak{a}M + N}{N}$ has the form

$$\frac{\mathfrak{a}M + N}{N} = \{x + N \mid x \in \mathfrak{a}M\}$$

as in (4).

Thus

$$\mathfrak{a}\left(\frac{M}{N}\right) = \mathfrak{a}M + N = \frac{\mathfrak{a}M + N}{N} \quad (5)$$

By the Collorary assumption, $M = \mathfrak{a}M + N$; quotient it by N :

$$\frac{M}{N} = \frac{\mathfrak{a}M + N}{N} \quad (6)$$

So, from (5) and (6):

$$\mathfrak{a}\left(\frac{M}{N}\right) = \mathfrak{a}M + N = \frac{\mathfrak{a}M + N}{N} = \frac{M}{N}$$

thus, $\mathfrak{a}(\frac{M}{N}) = \frac{M}{N}$.

By Nakayama's lemma AM.2.6, if $\mathfrak{a}(\frac{M}{N}) = \frac{M}{N} \xRightarrow{\text{implies}} \frac{M}{N} = 0$

Note that

$$\frac{M}{N} = \{m + N \mid m \in M\}$$

(the zero element in $\frac{M}{N}$ is the coset $N = 0 + N$)

Then, $\frac{M}{N} = 0$ means that the quotient has exactly one element, the zero coset N .

Thus, every coset $m + N$ equals the zero coset N , so $m - 0 \in N \implies m \in N$.

Hence every $m \in M$ lies in N , ie. $\forall m \in M, m \in N$.

So $M \subseteq N$. But notice that by the Corollary, we had $N \subseteq M$, therefore $M = N$.

Thus, if $M = \mathfrak{a}M + N \xRightarrow{\text{implies}} M = N$. □

Proposition AM.2.8. Let $x_i \forall i \in [n]$ be elements of M whose images $\frac{M}{mM}$ from a basis of this vector space. Then the x_i generate M .

Proof. Let N submodule M , generated by the x_i .

Then the composite map $N \longrightarrow M \longrightarrow \frac{M}{mM}$ maps N onto $\frac{M}{mM}$, hence $N + \mathfrak{a}M = M$, which by AM.2.7 implies $N = M$. □

2.3 Sequences

Definition R.2.9.a (Exact Sequence). Let a sequence of homomorphisms

$$L \xrightarrow{\alpha} M \xrightarrow{\beta} N$$

It is *exact* at M if $\text{im}(\alpha) = \ker(\beta)$.

ie. $\beta \circ \alpha = 0$ and α maps surjectively to $\ker(\beta)$.

Definition R.2.9.b (Short Exact Sequence (s.e.s.)).

$$0 \longrightarrow L \xrightarrow{\alpha} M \xrightarrow{\beta} N \longrightarrow 0$$

is exact $\iff L \subset M$ and $N = M/L$.

Properties:

- α injective
- β surjective
- $\alpha : L \implies \ker \beta$
- β induces $M/\alpha(L) \longrightarrow N$

Proposition R.2.10 (Split exact sequence). For the previous s.e.s., 3 equivalent conditions:

i. $\exists$ isomorphism $M \cong L \oplus N$, with

$$\begin{aligned}\alpha : m &\longmapsto (m, 0) \\ \beta : (m, n) &\longmapsto n\end{aligned}$$

ii. $\exists$ a *section* of β , that is, a map $s : N \longrightarrow M$ such that $\beta \circ s = id_N$

iii. $\exists$ a *retraction* of α , that is, a map $r : M \longrightarrow L$ such that $r \circ \alpha = id_L$

If all i, ii, iii are satisfied, it is a split exact sequence.

Proof. Intuitively, when a s.e.s. *splits* it means that the middle module M is the direct sum of the other (outer) two modules, ie. $M = L \oplus N$.

(i to ii, iii) if $M \cong L \oplus N$ such that $\alpha : m \longmapsto (m, 0)$, $\beta : s(m, n) \longmapsto n$, we can define the maps

for ii:

$$\begin{aligned}s : N &\longrightarrow L \oplus N \\ s(n) &\longmapsto (0, n)\end{aligned}$$

Then $\beta(s(n)) = \beta(0, n)$, so $\beta \circ s = id_N$.

for iii:

$$\begin{aligned}r : L \oplus N &\longrightarrow L \\ r(m, n) &\longmapsto m\end{aligned}$$

Then $r(\alpha(m)) = r(m, 0)$, so $r \circ \alpha = id_L$.

(ii to i) assume $s : N \longrightarrow M$ such that $\beta \circ s = id_M$

Want to show $M \cong im(\alpha) \oplus im(s)$.

$\forall m \in M$, consider $m - s(\beta(m))$, apply β to it:

$$\beta(m - s(\beta(m))) = \beta(m) - (\beta \circ s)(\beta(m)) = \beta(m) - \beta(m) = 0$$

Since $ker(\beta) = im(\alpha)$, $\exists ! l \in L$ such that $\alpha(l) = m - s(\beta(m))$.

Thus $m = \alpha(l) + s(\beta(m))$.

Now, suppose $x \in im(\alpha) \cap im(s)$, then $x = \alpha(l) = s(n)$, apply β to it:

$$\beta(\alpha(l)) = \beta(s(n)) \implies 0 = n.$$

If $n = 0$, then $s(n) = 0$, so the intersection is $\{0\}$.

Define

$$\begin{aligned}\phi : L \oplus N &\longrightarrow M \\ \phi(l, n) &\longmapsto \alpha(l) + s(n)\end{aligned}$$

This isomorphism satisfies the required conditions.

(iii to i) similar to the previous one.

Overview:

$$0 \longrightarrow L \xrightleftharpoons[r]{\alpha} \xrightarrow[\cong]{M} L \oplus N \xleftarrow[s]{\beta} N \longrightarrow 0$$

$$\begin{aligned}\alpha : l &\longmapsto (l, 0) \\ r : (m, n) &\longmapsto m \\ \alpha \circ r &= id_L \\ \beta : (l, n) &\longmapsto n \\ s : n &\longmapsto (0, n) \\ \beta \circ s &= id_N\end{aligned}$$

□

3 Noetherian rings (and modules)

Definition (Ascending Chain Condition). A partially ordered set Σ has the *ascending chain condition* (a.c.c.) if every chain

$$s_1 \leq s_2 \leq \dots \leq s_k \leq \dots$$

eventually breaks off, that is, $s_k = s_{k+1} = \dots$ for some k .

$\implies$ Σ has the a.c.c. iff every non-empty subset $S \subset \Sigma$ has a maximal element.

if $\neq S \subset \Sigma$ does not have a maximal element, choose $s_1 \in S$, and for each s_k , an element s_{k+1} with $s_k < s_{k+1}$, thus contradicting the a.c.c.

3.1 Noetherian rings and modules

Definition R.3.2 (Noetherian ring). Let A a ring; 3 equivalent conditions:

- i. the set Σ of ideals of A has the a.c.c.; in other words, every increasing chain of ideals

$$I_1 \subset I_2 \subset \dots \subset I_k \subset \dots$$

eventually stops, that is $I_k = I_{k+1} = \dots$ for some k .

- ii. every nonempty set S of ideals has a maximal element
- iii. every ideal $I \subset A$ is finitely generated

If these conditions hold, then A is *Noetherian*.

Proof. TODO □

Definition R.3.4.D (Noetherian modules). An A -module M is Noetherian if the submodules of M have the a.c.c., that is, any increasing chain

$$M_1 \subset M_2 \subset \dots \subset M_k \subset \dots$$

of submodules eventually stops.

As in with rings, it is equivalent to say that

- i. any nonempty set of submodules of M has a maximal element
- ii. every submodule of M is finite

Proposition R.3.4.P. Let $0 \longrightarrow L \xrightarrow{\alpha} M \xrightarrow{\beta} N \longrightarrow 0$ be a s.e.s. (split exact sequence, R.2.10).

Then, M is Noetherian $\iff L$ and N are Noetherian.

Proof. $\implies$: trivial, since ascending chains of submodules in L and N correspond one-to-one to certain chains in M .

$\impliedby$: suppose $M_1 \subset M_2 \subset \dots \subset M_k \subset \dots$ is an increasing chain of submodules of M .

Then identifying $\alpha(L)$ with L and taking intersection gives a chain

$$L \cap M_1 \subset L \cap M_2 \subset \dots \subset L \cap M_k \subset \dots$$

of submodules of L , and applying β gives a chain

$$\beta(M_1) \subset \beta(M_2) \subset \dots \subset \beta(M_k) \subset \dots$$

of submodules of N .

Each of these two chains eventually stop, by the assumption on L and N , so that we only need to prove the following lemma which completes the proof. □

Lemma R.3.4.L. for submodules $M_1 \subset M_2 \subset M$,

$$L \cap M_1 = L \cap M_2 \text{ and } \beta(M_1) = \beta(M_2) \implies M_1 = M_2$$

Proof. if $m \in M_2$, then $\beta(m) \in \beta(M_1) = \beta(M_2)$, so that there is an $n \in M_1$ such that $\beta(m) = \beta(n)$.

Then $\beta(m - n) = 0$, so that

$$m - n \in M_2 \cap \ker(\beta) = M_1 \cap \ker(\beta)$$

Hence $m \in M_1$, thus $M_1 = M_2$. □

Corollary R.3.5 (Properties of Noetherian modules). i. if $\forall i \in [r]$, M_i are Noetherian modules, then $\bigoplus_{i=1}^r M_i$ is Noetherian.

- ii. if A a Noetherian ring, then an A -module M is Noetherian iff it is finite over A .
- iii. if A a Noetherian ring, M a finite module, then any submodule $N \subset M$ is again finite.
- iv. if A a Noetherian ring, and $\psi : A \longrightarrow B$ a ring homomorphism such that B is a finite A -module, then B is a Noetherian ring.

Proof. i. a direct sum $M_1 \oplus M_2$ is a particular case of an exact sequence.

Then, Proposition R.3.4.P proves this statement when $r = 2$. The case $r > 2$ follows by induction.

- ii. if M finite, then $\exists$ surjective homomorphism

$$A^r \longrightarrow M \longrightarrow 0$$

for some r , so that M is a quotient

$$M \cong A^r/N$$

for some submodule $N \subset A^r$.

A^r is a Noetherian module by i., so M is Noetherian due Proposition R.3.4.P.

Conversely, M Noetherian implies M finite.

item as in previous implications:

M finite and A Noetherian $\implies M$ is Noetherian,
 $\implies$ since $N \subseteq M$, then N is Noetherian too
 $\implies$ which implies that N is a finite A -module.

- iii. B is Noetherian as an A -module; but ideals of B are submodules of B as an A -submodule, so that B is a Noetherian ring.

□

3.2 Hilbert basis

Theorem R.3.6 (Hilbert basis theorem). If A a Noetherian ring, then so is the polynomial ring $A[x]$.

- ie. Every ideal in a polynomial ring over a field has a finite set of generators (is fingen).

Proof. Prove that any ideal $I \subset A[x]$ is fingen.

Define auxiliary sets $J_n \subset A$ by

$$J_n = \{a \in A \mid \exists f \in I \text{ s.th. } f = ax^n + b_{n-1}x^{n-1} + \dots b_0\}$$

ie. J_n is the set of leading coefficients of I of degree n .

J_n is an ideal, since I is an ideal.

$J_n \subset J_{n+1}$, since for $f \in I$ also $xf \in I$.

Therefore $J_1 \subset J_2 \subset \dots \subset J_k \subset \dots$ is an increasing chain of ideals.

Using the assumption that A is Noetherian, deduce that $J_n = J_{n+1}$ for some n .

For each $m \leq n$, $J_m \subset A$ is finiten, ie.

$$J_m = (a_{m,1}, \dots, a_{m,r_m})$$

By definition of J_m , for each $a_{m,j}$ with $1 \leq j \leq r_m$,

$\exists$ a polynomial $f_{m,j} \in I$ of degree m having the leading coefficient $a_{m,j}$.

$$\implies \{f_{m,j}\}_{m < n; 1 \leq j \leq r_m}$$

the set of elements of I .

Claim: this finite set $(\{f_{m,j}\})$ generates I .

$\forall f \in I$, if $\deg f = m$, then its leading coefficient is $a \in J_m$,

hence if $m \geq n$, then $a \in J_m = J_n$, so that

$$a = \sum b_i a_{n,i} \text{ with } b_i \in A$$

and

$$f - \sum b_i X^{m-n} \cdot f_{n,i}$$

has degree $< m$.

Similarly, if $m \leq n$, then $a \in J_m$, so that

$$a = \sum b_i a_{m,i} \text{ with } b_i \in A$$

and

$$f - \sum b_i f_{n,i}$$

has degree $< m$.

By induction on m , f can be written as a linear combination of finitely many elements.

Thus, any ideal of $A[x]$ is finitely generated. $\square$

Corollary R.3.6.C. if A a Noetherian ring, and $\psi : A \longrightarrow B$ a ring homomorphism such that B is a finiten extension ring of $\psi(A)$, then B is Noetherian.

In particular, any finiten algebra over $\mathbb{Z}$ or over a field K is Noetherian.

Proof. the assumption is that B is a quotient of a polynomial ring,

$$B \cong A[x_1, \dots, x_n]/I$$

for some ideal I .

By the Hilbert basis theorem R.3.6 and induction,

A being Noetherian implies that $A[x_1, \dots, x_n]$ is Noetherian.

And by Corollary R.3.5(iv),

$A[x_1, \dots, x_n]$ being Noetherian implies that $A[x_1, \dots, x_n]/I$ is Noetherian. $\square$

4 Finite ring extensions and Noether normalization

4.1 A-algebras and integral domains

Definition (A-algebra / k-algebra). An A -algebra is a ring B with a ring homomorphism $\psi : A \longrightarrow B$.

B is an A -module with multiplication defined by $\psi(a) \cdot b$ ($a \in A, b \in B$).

When $A \subset B$, B is an extension ring of A ; denoted $\psi(A) = A' \subset B$.

Definition R.4.1. Let B be an A -algebra.

- i. B is a *finite* A -algebra (*finite over* A) if it is finite as an A -module.
- ii. $y \in B$ is *integral over* A if $\exists$ a monic polynomial

$$f(Y) = Y^n + a_{n-1}Y^{n-1} + \dots + a_0 \in A'[Y]$$

such that $f(y) = 0$:

$$f(y) = y^n + a_{n-1}y^{n-1} + \dots + a_0 = 0$$

The algebra B is *integral over* A if $\forall b \in B$ b is integral.

Proposition R.4.2. Let $\psi : A \longrightarrow B$ be an A -algebra, and $y \in B$. Three equivalent conditions:

- i. y is integral over A
- ii. subring $A'[y] \subset B$ generated by $A' = \psi(A)$ and y is finite over A
- iii. $\exists$ an A -subalgebra $C \subset B$ such that $A'[y] \subset C$ and C is finite over A

Notes: A' is the image of A in B , ie. $A' = \psi(A)$.
 $A'[y]$ is the smallest subring of B containing both coefficients from A and the element y .

Proof. .

(i to ii): since y integral over $A \implies$ by R.4.1 (ii), y satisfies

$$f(y) = y^n + a_{n-1}y^{n-1} + \dots + a_0 = 0$$

So any power y^k ($k \geq n$) can be expressed in terms of $\{1, y, y^2, \dots, y^{n-1}\}$.

Thus the set $\{1, y, y^2, \dots, y^{n-1}\}$ spans $A'[y]$ as an A -module.

(iii to i): since $A'[y] \subset C \implies y \in C$
since C finite over A , C has finite generators $\{c_1, \dots, c_n\}$ such that $C = A \cdot c_1 + A \cdot c_2 + \dots + A \cdot c_n$

Thus $y \cdot c_i \in C$,

$$y \cdot c_i = \sum_{j=1}^n a_{ij} c_j$$

with $a_{ij} \in A$.

By the Cayley-Hamilton theorem (AM.2.4),

$$y^n + a_{n-1}y^{n-1} + \dots + a_1y + a_0 = 0$$

Therefore, y is integral (by R.4.1 (ii)).

□

Proposition R.4.3 (Tower Laws). Let B be an A -algebra.

- a. Transitivity of finiteness: if $A \subset B \subset C$ are extension rings such that C is a finite B -algebra and B a finite A -algebra, then C is finite over A .
- b. Finiteness of generated algebras: if $y_1, \dots, y_m \in B$ are integral over A , then $A[y_1, \dots, y_m]$ is finite over A .
In particular, every $f \in A[y_1, \dots, y_m]$ is integral over A .
- c. Transitivity of integrality: if $A \subset B \subset C$ with C integral over B , and B integral over A , then C is integral over A .
- d. Integral closure as a subring: the subset

$$\tilde{A} = \{y \in B \mid y \text{ is integral over } A\} \subset B$$

is a subring of B .

Moreover, if $y \in B$ is integral over $\tilde{A}$ then $y \in \tilde{A}$, so that $\tilde{\tilde{A}} = \tilde{A}$.

Proof. .

- a. if $\{\beta_1, \dots, \beta_n\}$ generate B as an A -module and $\{\gamma_1, \dots, \gamma_m\}$ generate C as an B -module,
then the set of products $\{\beta_i \gamma_j\}$ generates C as an A -module.
Since there are $n \times m$ generators (ie. finite), C is finite over A .
- b. proof by induction:
base case: if y_1 integral over $A \implies$ it satisfies a monic polynomial.

Thus $A[y_1]$ is generated as an A -module by $\{1, y_1, y_1^2, \dots, y_1^{n-1}\}$, making it a finite A -algebra.

inductive step: let $R_k = A[y_1, \dots, y_k]$. Assume R_k is finite over A .

Since y_{k+1} is integral over $A \implies$ it is also integral over R_k .

Thus $R_{k+1} = R_k[y_{k+1}]$ is finite over R_k .

Applying part (a) (transitivity of finiteness), if R_{k+1} is finite over R_k and R_k finite over A , then R_{k+1} is finite over A .

Consequence: since any $f \in A[y_1, \dots, y_m]$ belongs to a finite A -algebra, f must be integral over A (since an element is integral iff it is contained in a finite extension).

c. let $x \in C$, since x integral over B , it satisfies:

$$x^n + b_{n-1}x^{n-1} + \dots + b_1x + b_0 = 0, \quad b_i \in B$$

Let $B'' = A[b_0, b_1, \dots, b_{n-1}]$. Since each $b_i \in B$ and B is integral over $A \implies$ each b_i is integral over A .

Since all b_i are integral over $B' \implies B'[x]$ is a finite B' -algebra.

By part (a) (transitivity of finiteness), $B'[x]$ is a finite A -algebra.

Therefore, x is integral over A .

d. I. subring:

let $x, y \in \tilde{A}$. Want to show $x + y, xy \in \tilde{A}$:

by part (b), the algebra $A[x, y]$ is finite over A .

Since $x + y, xy \in A[x, y]$, they are integral over A .

Thus $x + y, xy \in \tilde{A}$, since $\tilde{A} = \{b \in B \mid b \text{ integral over } A\}$.

II. idempotence

let $z \in B$ be integral over $\tilde{A}$

we have a chain $A \subseteq \tilde{A} \subseteq \tilde{A}[x]$.

By definition, $\tilde{A}$ is integral over A , and z is integral over $\tilde{A}$

thus by part (c), z is integral over A .

Therefore, $z \in \tilde{A}$.

□

Lemma 4.3.Aux (Integrality implies finiteness). If y integral over A then $A[y]$ is finite over A .

This extends on point (b) from the previous proposition R.4.3.

Proof. Suppose y is integral over A . By definition $\exists f \in A[T]$, with f monic, such that $f(y) = 0$. (here T is a formal variable).

Let $\deg(f) = d$, so that for $f(y) = 0$ we have

$$y^d + a_{d-1}y^{d-1} + \dots + a_1y + a_0 = 0 \quad a_i \in A$$

Since it is monic (leading coefficient is 1), we can rearrange it to isolate the highest power:

$$y^d = -(a_{d-1}y^{d-1} + \dots + a_1y + a_0) \quad (7)$$

Thus y^d can be written using lower powers of y with coefficients in A .

Consider any element $p \in A[y]$, $p = c_my^m + c_{m-1}y^{m-1} + \dots + c_0$.

if $m < d$, leave it as it is.

if $m \geq d$, use the monic equation (7) to replace y^d with lower powers.

Repeating this process, can reduce any power of y down to a linear combination of $\{1, y, y^2, \dots, y^{d-1}\}$.

Thus every element in $A[y]$ can be expressed as

$$\lambda_{d-1}y^{d-1} + \dots + \lambda_2y^2 + \lambda_1y + \lambda_0 \cdot 1 \quad \lambda_i \in A$$

Henceforth, the set $\{1, y, y^2, \dots, y^{d-1}\}$ generates $A[y]$ as a finite A -module. $\square$

Definition 4.4 (Integral closure). Given the ring $\tilde{A}$ from R.4.3.(d), ie. $\tilde{A} = \{y \in B \mid y \text{ integral over } A\} \subset B$, $\tilde{A}$ is the *integral closure* of A in B .

If $A = \tilde{A}$, then A is *integrally closed* in B .

An integral domain A is *normal* if it is *integrally closed in its field of fractions*, that is if

$$A = \tilde{A} \subset K = \text{Frac}(A)$$

For any integral domain A , the integral closure of A in its field of fractions $K = \text{Frac}(A)$ is also called the *normalization* of A .

4.2 Noether normalization

Definition 4.6 (Algebraically independent). $y_1, \dots, y_n \in A$ are *algebraically independent* over K if the natural surjection $K[Y_1, \dots, Y_n] \twoheadrightarrow K[y_1, \dots, y_n]$ is an isomorphism.

$\implies \nexists F(y_1, \dots, y_n) = 0$ (F nonzero) with coefficients in K .

Recall: a K -algebra A is *fingen* over K if $A = K[y_1, \dots, y_n]$ for some finite set $y_1, \dots, y_n$.

Lemma R.4.6.L. Let $A = K[y_1, \dots, y_n]$ and $0 \neq F \in K[Y_1, \dots, Y_n]$ such that $F(y_1, \dots, y_n) = 0$.

Then $\exists y_1^*, \dots, y_{n-1}^* \in A$ such that y_n is integral over

$$A^* = K[y_1^*, \dots, y_{n-1}^*] \quad \text{and} \quad A = A^*[y_n]$$

Proof. Set $y_i^* = y_i - y_n^{r_i}$ for $i \in [n-1]$ and $r_1, \dots, r_{n-1} \geq 1 \in \mathbb{Z}$.

(ie. $y_i = y_i^* + y_n^{r_i}$)

Define $G \in A$ by

$$G(y_1^*, \dots, y_{n-1}^*, y_n) = F(y_i^* + y_n^{r_i}, y_n) = 0$$

viewed as a relation for y_n over $K[y_1^*, \dots, y_{n-1}^*]$.

Since F is a polynomial in $y_1, \dots, y_{n-1}^*$, can write it as a sum of monomials

$$F = \sum_m a_m y^m = \sum_m a_m \prod y_i^{m_i}$$

where $m = (m_1, \dots, m_n)$ and each $a_m \neq 0$.

Therefore,

$$G = \sum a_m \prod (y_i^* + y_n^{r_i})^{m_i}$$

which when expanding out, each summand $a_m \prod (y_i^* + y_n^{r_i})^{m_i}$ has a unique term of highest order in y_n , namely $a_m y_n^{(\sum r_i m_i)}$.

Suppose we can arrange so that

$$m \neq m' \implies \sum r_i m_i \neq \sum r_i m'_i$$

Then $\max\{\sum r_i m_i \mid m \text{ s.th. } a_m \neq 0\}$ is achieved in only one summand, so that there is no cancellation; thus the highest order term in G is $a_m y_n^{(\sum r_i m_i)}$ (ie. a_m times a pure power of y_n). $\square$

Theorem R.4.6 (Noether normalization lemma). Let K a field, A a fingen K -algebra.

Then $\exists z_1, \dots, z_m \in A$ such that

- i. $z_1, \dots, z_m$ are algebraically independent over K
- ii. A is finite over $B = K[z_1, \dots, z_m]$

That is, a fingen extension $K \subset A$ can be written as a composite

$$K \subset B = K[z_1, \dots, z_m] \subset A$$

where $K \subset B$ is a polynomial extension, and $B \subset A$ is finite.

Proof. Let $y_1, \dots, y_n$ be generators of $A = K[y_1, \dots, y_n]$.

if $n = 0$, nothing to prove since A is generated by 0 elements $\implies A = K$, and K is finite.

if $n > 0$ we have two cases:

- $y_1, \dots, y_n$ are algebraically independent over K , then by definition 4.6 $A \cong K[y_1, \dots, y_n]$, so that A is a finite module over itself, with $m = n$.

- $y_1, \dots, y_n$ are algebraically dependent over K ,

$$\exists 0 \neq f \in K[y_1, \dots, y_n] \text{ s.th } f(y_1, \dots, y_n) = 0$$

Want f to be *monic*, so that y_n is integral over new defined variables $y_1^*, \dots, y_{n-1}^*$. In other words, want some polynomial like

$$y_n^d + a_{d-1}y_n^{d-1} + \dots + a_1y_n + a_0 = 0 \quad a_i \in K[y_1^*, \dots, y_{n-1}^*]$$

ie. monic, so that by definition (R.4.1), y_n is integral over $K[y_1^*, \dots, y_{n-1}^*]$.
 $\longrightarrow$ Change variables so that f becomes monic in one of the variables (y_n); this allows to express one generator (y_n) as an integral element over the others.

Following from Lemma R.4.6.L, define the new variables $y_1^*, \dots, y_{n-1}^* \in A$ such that y_n is integral over

$$A^* = K[y_1^*, \dots, y_{n-1}^*], \text{ and } A = A^*[y_n]$$

Setting $y_i^* = y_i - y_n^{r_i}$, so that $y_i = y_i^* + y_n^{r_i} \forall i \in [n-1]$, $r_1, \dots, r_{n-1} \geq 1 \in \mathbb{Z}$.
 Use those new variables at $f(y_1, \dots, y_n) = 0$:

$$f(y_1^* + y_n^{r_1}, y_2^* + y_n^{r_2}, \dots, y_{n-1}^* + y_n^{r_{n-1}}, y_n) = 0$$

Then the highest power of y_n in each term of f will look like $y_n^{(\sum a_i r_i)}$, and with r_i growing fast enough we ensure that each monomial in f produces a unique power of y_n .

Then we have $c \cdot y_n^D + (\text{terms with lower powers of } y_n) = 0$ with $c \in K \setminus \{0\}$. So that dividing by c we get the shape $y_n^D + \dots = 0$, thus y_n is integral over $A^* = K[y_1^*, \dots, y_{n-1}^*]$.

Induction:

Since y_n integral over $A^* \implies A = A^*[y_n]$ is finite over $A^* = K[y_1^*, \dots, y_{n-1}^*]$ (by 4.3.Aux).

By inductive hypothesis on A^* , $\exists z_1, \dots, z_m \in A^*$ algebraically independent over K and with A^* finite over $B = K[z_1, \dots, z_m]$.

Therefore, each step of $B \subset A^* \subset A^*[y_n] = A$ is finite, and A is finite over B as required. $\square$

Example . $A = K[X, Y]/(XY - 1)$. Y is algebraic over $K[X]$, but not integral over $K[Y]$.

This corresponds to the fact that the hyperbola $XY = 1$ has the line $X = 0$ as an asymptotic line (so that its projection to the X -axis misses a root over $X = 0$).

Take $X' = X - \epsilon Y$ as the element of A instead of X ; then the relation becomes $(X' + \epsilon Y)Y = 1$, monic in Y if $\epsilon \neq 0$.

This corresponds geometrically to tilting the hyperbola a little before projecting, so that no longer has a vertical asymptotic line.

4.3 Weak Nullstellensatz

Proposition R.4.9. let $A \subset B$ be an integral extension of integral domain, then A is a field $\iff B$ is a field.

Proof. $\implies$:

let $0 \neq x \in B$, then $\exists x^n + a_{n-1}x^{n-1} + \dots + a_0 = 0 \quad a_i \in A$, monic.

Since A is a field, $\exists$ inverse, observe that:

$$\begin{aligned} x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0 &= 0 \\ x(x^{n-1} + a_{n-1}x^{n-2} + \dots + a_1) &= -a_0 \\ -a_0^{-1}(x^{n-1} + a_{n-1}x^{n-2} + \dots + a_1) &= x^{-1} \in B \end{aligned}$$

thus there exists inverse in B , so B is a field too.

$\impliedby$:

if B is a field and $0 \neq x \in A$, then $x^{-1} \in B$, so x^{-1} is integral over A .

So there is a relation of the form

$$(x^{-1})^n + a_{n-1}(x^{-1})^{n-1} + \dots + a_0 = 0$$

Therefore

$$\begin{aligned} (x^{-1})^n + a_{n-1}(x^{-1})^{n-1} + \dots + a_0 &= 0 \\ (x^{-1})^n &= -a_{n-1}(x^{-1})^{n-1} - \dots - a_0 \\ x^{-n} &= -a_{n-1}x^{-n+1} - \dots - a_0 \quad (\text{mult by } x^{n-1}) \\ x^{-n+(n-1)} &= -a_{n-1}x^{-n+1+(n-1)} - \dots - a_0x^{n-1} \\ x^{-1} &= -a_{n-1} - \dots - a_0x^{n-1} \in A \end{aligned}$$

thus there exists inverse in A , so A is a field too. $\square$

Theorem 4.10.prev (Zariski's lemma). k a field, L fingen k -algebra and a field. Then L is a finite algebraic extension of k .

If k is algebraically closed, then $L = k$.

From Zariski's lemma we can see that:

let maximal ideal $m \subset k[X_1, \dots, X_n]$, then $k[X_1, \dots, X_n]/m$ is a (quotient) field.

It's a fingen k -algebra, so by Zariski's lemma, $k[X_1, \dots, X_n]/m$ is algebraic over k .

Since $k[X_1, \dots, X_n]/m$ is algebraically closed, it must equal k ; ie.

$$\frac{k[X_1, \dots, X_n]}{m} = k$$

Since $k[X_1, \dots, X_n]$ is a k -algebra, we have the natural map

$$\psi : k \longrightarrow \frac{k[X_1, \dots, X_n]}{m}$$

so that $\forall x_i \in k$, $\psi(x_i) = a_i \in k$ (its image in the quotient field $k[X_1, \dots, X_n]/m$ must be an element of k).

Then, by definition of a quotient ring, $\psi(x_i)$ is the coset $\psi(x_i) = X_i + m$. Thus,

$$\begin{aligned} a_i &= X_i + m \\ \implies X_i - a_i &\in m \end{aligned}$$

The ideal generated by these terms is a subset of m :

$$J = (X_1 - a_1, \dots, X_n - a_n) \subseteq m$$

Thus $m = (X_1 - a_1, \dots, X_n - a_n)$ (shown also at 5.2).

Then: For k algebraically closed, every maximal ideal of $k[X_1, \dots, X_n]$ has the form $(X_1 - a_1, \dots, X_n - a_n)$ for some $a_i \in k$.

Equivalently, $V(I) = \emptyset \iff 1 \in I$.

Theorem R.4.10 (Weak Nullstellensatz - Zariski's lemma). let k a field, K a k -algebra which

1. is finitely generated as a k -algebra
2. is a field

Then K is algebraic over k , so that $k \subset K$ is a finite field extension. That is, $[K : k] < \infty$.

Proof. $K = k[z_1, \dots, z_m]$ a field; want to show that K is algebraic over k .

Since K is a finitely generated k -algebra, by Noether normalization lemma (R.4.6), $\exists z_1, \dots, z_m \in K$ such that

- are algebraically independent
- K is integral over the polynomial ring $A = k[z_1, \dots, z_m]$ (which by 4.3.Aux is finite)

Now we're at the situation of R.4.9:

$A \subset K$ is integral, K is a field $\implies$ therefore A is a field.

Since $z_1, \dots, z_m \in K$ are algebraically independent,

$\implies A = k[z_1, \dots, z_m]$ is a polynomial ring in m indeterminates, and this is a field only if $m = 0$

(since in $k[z_1]$ the element z_1 is not invertible, since $1/z_1$ is a rational function, not a polynomial).

So $A = k$; which by Noether normalization we saw that K is integral over $A = k$, and by 4.3.Aux that it is finite, thus K is finite over k , ie. $[K : k] < \infty$, and K is algebraic over k . $\square$

5 Nullstellensatz

5.1 Corollary from Weak Nullstellensatz

Note: for k a field, $k[X_1, \dots, X_n]$, m maximal ideal; the residue field $K = k[X_1, \dots, X_n]/m$ satisfies the Zariski's lemma (4.10.prev), thus K is a finite algebraic extension of k .

Corollary 5.2. k algebraically closed. Then every maximal ideal of $A = k[X_1, \dots, X_n]$ is of the form

$$m = (X_1 - a_1, \dots, X_n - a_n), \quad a_i \in k$$

The map $k[X_1, \dots, X_n] \rightarrow k[X_1, \dots, X_n]/m = k$ is the natural evaluation map $f(X_1, \dots, X_n) \mapsto f(a_1, \dots, a_n)$.

Thus

$$\begin{aligned} k^n &\longleftrightarrow m - \text{Spec } A \\ (a_1, \dots, a_n) &\longleftrightarrow f(a_1, \dots, a_n) \end{aligned}$$

Proof. let $m \subset k[X_1, \dots, X_n]$ be a maximal ideal.

By fundamental property of maximal ideals, $K = A/m$ is a field.

Since A is a finitely generated k -algebra (generated by $X_1, \dots, X_n$), then $K = A/m$ is also a finitely generated k -algebra, generated by residues $\psi(x_i) = x'_i = x_i + m$ (where $\psi : k \rightarrow A/m$).

By Zariski's lemma (4.10.prev), $K = A/m$ is algebraic over k .

Since by hypothesis k is algebraically closed, it has no proper algebraic extensions

$$\implies K = k \implies k \cong A/m.$$

Recall that since A is a k -algebra, we have the natural map $\psi : k \rightarrow A/m$, so that $\forall x_i \in k$, $\psi(x_i) = a_i \in k$ (its image in the quotient field A/m must be an element of k).

Also, by definition of a quotient ring, $\psi(x_i)$ is the coset $\psi(x_i) = X_i + m$. Thus,

$$\begin{aligned} a_i &= X_i + m \\ \implies X_i - a_i &\in m \end{aligned}$$

The ideal generated by these terms is a subset of m :

$$J = (X_1 - a_1, \dots, X_n - a_n) \subseteq m$$

Since J is the kernel of the evaluation map at point $(a_1, \dots, a_n)$, then J is a maximal ideal. Together with $J \subseteq m$, then we have $J = m$, ie.

$$m = (X_1 - a_1, \dots, X_n - a_n)$$

Let

$$\begin{aligned}\phi : k[X_1, \dots, X_n] &\longrightarrow k[X_1, \dots, X_n]/m \\ \phi : x_i &\longmapsto a_i\end{aligned}$$

Which is the natural evaluation map $f(X_1, \dots, X_n) \longrightarrow f(a_1, \dots, a_n)$.

Since ϕ is a k -algebra homomorphism, then $\forall f \in A$:

$$\phi(f(X_1, \dots, X_n)) = f(\phi(x_1), \dots, \phi(x_n)) = f(a_1, \dots, a_n)$$

Thus there is a one-to-one correspondence:

$$\begin{array}{ll} \text{points in } k^n & \longleftrightarrow m - \text{Spec } A \text{ (maximal ideals in } k[X_1, \dots, X_n]) \\ (a_1, \dots, a_n) & \longleftrightarrow (X_1 - a_1, \dots, X_n - a_n) \end{array} \quad \square$$

5.2 Variety

Definition 5.3 (Variety). A *variety* $V \subset k^n$:

$$V = V(J) = \{P = (a_1, \dots, a_n) \in k^n \mid f(P) = 0 \forall f \in J\}$$

$\rightarrow V$ is defined by $f_1(P) = \dots = f_m(P) = 0$

$\rightarrow V$ is defined as the simultaneous solutions of a number of polynomial equations.

Proposition 5.3. k an algebraically closed field, and $A = k[X_1, \dots, X_n]$ a finitely generated k -algebra of the form $A = k[X_1, \dots, X_n]/J$, where J is an ideal of $k[X_1, \dots, X_n]$. (notation: $x_i = X_i \pmod{J}$)

Then every maximal ideal of A is of the form

$$(x_1 - a_1, \dots, x_n - a_n)$$

for some point $(a_1, \dots, a_n) \in V(J)$.

Therefore, $\exists$ a one-to-one correspondence

$$\begin{aligned} V(X) &\longleftrightarrow m - \text{Spec } A \\ \text{given by } (a_1, \dots, a_n) &\longleftrightarrow (x_1 - a_1, \dots, x_n - a_n) \end{aligned}$$

Proof. the ideals of A are given by ideals of $k[X_1, \dots, X_n]$ containing J , since for $Q = R/I$, $\exists$ one-to-one correspondence between ideals of Q and ideals of R that contain I , ie.

$$\begin{aligned} m \subset A &\longleftrightarrow m' \subseteq k[X_1, \dots, X_n] \\ \text{s.th. } J &\subseteq m' \end{aligned}$$

Thus, every maximal ideal of A is of the form

$$\underbrace{(x_1 - a_1, \dots, x_n - a_n)}_{i.} \text{ such that } \underbrace{J \subset (X_1 - a_1, \dots, X_n - a_n)}_{ii.}$$

- i. Since k is algebraically closed, the maximal ideals of $k[X_1, \dots, X_n]$ look like $m = (X_1 - a_1, \dots, X_n - a_n)$ for some point $(a_1, \dots, a_n) \in k^n$.
Which when projected to the quotient ring A , $X_i \mapsto x_i$ (residue class), giving $(x_1 - a_1, \dots, x_n - a_n)$.
- ii. for m to exist in A , the corresponding m' must contain J ; since if it didn't contain J it wouldn't "survive" the quotient process.

However, since $(X_1 - a_1, \dots, X_n - a_n)$ is the kernel of the evaluation map $f \mapsto f(a_1, \dots, a_n)$

$\implies$ means that $m' = (X_1 - a_1, \dots, X_n - a_n)$ consists of all polynomials that vanish at point $P = (a_1, \dots, a_n)$.

If $J \subseteq m'$, then $\forall f \in J$ must vanish at P .

By definition, the set of points where all polynomials in J vanish is the *variety*, $V(J)$.

Thus,

every maximal ideal in A corresponds to a point $(a_1, \dots, a_n) \in k^n$, ie.

$$m - \text{Spec } A \longleftrightarrow k^n$$

The condition that the ideal belongs to the quotient ring $A = k[X_1, \dots, X_n]/J$ forces that point to lie in $V(J)$, so

$$\begin{aligned} m - \text{Spec } A &\longleftrightarrow V(J) \\ \text{maximal spectrum} &\longleftrightarrow \text{variety} \end{aligned}$$

□

Proposition 5.5 (Correspondences V and I). A variety $X \subset k^n$ is by definition $X = V(J)$ (J an ideal of $k[X_1, \dots, X_n]$).

So V gives a map:

$$\{\text{ideals of } k[X_1, \dots, X_n]\} \xrightarrow{V} \{\text{subsets } X \text{ of } k^n\}$$

correspondence going the other way:

$$\{\text{subsets } X \text{ of } k^n\} \xrightarrow{I} \{\text{ideals of } k[X_1, \dots, X_n]\}$$

defined by taking a subset $X \subset k^n$ into the ideal

$$I(X) = \{f \in k[X_1, \dots, X_n] \mid f(P) = 0 \ \forall P \in X\}$$

V , I satisfy reverse inclusions:

$$J \subset J' \implies V(J) \supset V(J') \quad \text{and} \quad X \subset Y \implies I(X) \supset I(Y)$$

5.3 Nullstellensatz

Theorem 5.6 (Nullstellensatz). Let k algebraically closed field.

- a. if $J \subsetneq k[X_1, \dots, X_n]$ then $V(J) \neq \emptyset$
b. $I(V(J)) = \text{rad}J$, in other words, for $f \in k[X_1, \dots, X_n]$,

$$f(P) = 0 \ \forall P \in V \iff f^n \in J \text{ for some } n.$$

Proof. a. if $J \subsetneq k[X_1, \dots, X_n]$ then $V(J) \neq \emptyset$:
Let $m \subset k[X_1, \dots, X_n]$ be a maximal ideal.

Now,

- since m maximal, $L = k[X_1, \dots, X_n]/m$ is a field.
- since $k[X_1, \dots, X_n]$ is a fingen k -algebra, L is a fingen k -algebra.

$\implies$ thus L is a finite field extension of k .

(Recall: if k algebraically closed and L a fingen field extension of k , then $L \cong k$.)

Therefore,

$$L = \frac{k[X_1, \dots, X_n]}{m} \cong k$$

Since $k[X_1, \dots, X_n]$ is a k -algebra, $\exists$ a surjective homomorphism

$$\psi : k[X_1, \dots, X_n] \longrightarrow k$$

Let $a_i = \psi(x_i)$. Then $x_i - a_i \in \ker(\psi) = m \ \forall i \in [n]$.

Since the ideal $(X_1 - a_1, \dots, X_n - a_n)$ is maximal and contained in m , they must be equal, ie. $m = (X_1 - a_1, \dots, X_n - a_n)$. (as in 5.2)

Therefore, $P = (a_1, \dots, a_n) \in k^n$ is a zero for every polynomial in m ,

$$f \in m \iff f(P) = 0$$

Since $J \subseteq m$, P is also a zero for every polynomial in J , ie. every element of J vanishes at P .

$\implies$ therefore $P \in V(J)$, and thus $V(J) \neq \emptyset$.

- b. $I(V(J)) = \text{rad}J$:

$$I(V(J)) = \text{rad}J$$

vanishing ideal of a variety = radical of the ideal defining the variety

where $\text{rad } J = \{f \in R \mid f^n \in J \text{ for some } n > 0\}$.

Want to show that if a polynomial vanishes at all points where $g_1, \dots, g_m$ vanish, then $f \in \text{rad}(g_1, \dots, g_m)$.

Consider the ring $k[X_1, \dots, X_n, Y]$ and the ideal J' generated by $\{g_1, \dots, g_m, 1 - Yf\}$

Suppose there is a point $(a_1, \dots, a_n, a_{n+1})$ that is a zero of J' . ie.

$$\exists (a_1, \dots, a_n, a_{n+1}) \in V(J')$$

Since $g_i(a) = 0$, our hypothesis says $f(a) = 0$. However, the last generator $(1 - Yf)$ requires

$$1 - a_{n+1}f(a) = 0 \implies \text{implies } 1 - a_{n+1} \cdot 0 = 0 \implies 1 - 0 = 0$$

a contradiction.

Therefore, $V(J') = \emptyset$.

Since $V(J') = \emptyset$, by the Weak Nullstellensatz/Zariski (4.10.prev),
if $V(J') = \emptyset$ then $J' = (1)$, so $1 \in J' = (1)$.

Every element in an ideal is a linear combination of its generators: J' is generated by $\{g_1, \dots, g_m, 1 - Yf\}$

$$\implies \forall j \in J', \quad j = \left(\sum (\text{polynomial}) g_i \right) + (\text{polynomial}) \cdot (1 - Yf)$$

which, since $1 \in J'$,

$$1 = \left(\sum_{i=1}^m p_i(X, Y) g_i(X) \right) + q(X, Y) \cdot (1 - Yf(X))$$

substitute $Y = 1/f$,

$$1 = \left(\sum_{i=1}^m p_i(X, \frac{1}{f}) g_i(X) \right) + q(X, \frac{1}{f}) \cdot \underbrace{\left(1 - \frac{1}{f} f(X) \right)}_0$$

thus

$$1 = \sum_{i=1}^m p_i(X, \frac{1}{f}) g_i(X)$$

multiply by f^n ,

$$f^n = \sum_{i=1}^m A_i(X) g_i(X)$$

thus f^n is a linear combination of g_i .

Thus $f^n \in J$, so $f \in \text{rad } J$.

□

5.4 Irreducible varieties

Definition 5.7 (Irreducible variety). a variety $X \subset k^n$ is *irreducible* if it is nonempty and not the union of two proper subvarieties; that is, if

$$X = X_1 \cup X_2 \text{ for varieties } X_1, X_2 \implies X = X_1 \text{ or } X_2$$

Proposition 5.7. a variety X is irreducible iff $I(X)$ is prime.

Proof. set $I = I(X)$.

if I not prime, then $f, g \in A \setminus I$ be such that $fg \in I$.

Define new ideals

$$J_1 = (I, f) \text{ and } J_2 = (I, g)$$

Then, since $f \notin I(X)$, it follows that $V(J_1) \subsetneq X$

$\implies$ so $X = V(J_1) \cup V(J_2)$ is reducible.

The converse is similar. □

Corollary 5.8. let k algebraically closed field. Then V and I induce one-to-one correspondences

$$\begin{aligned} \{\text{radical ideals } J \text{ of } k[X_1, \dots, X_n]\} &\longleftrightarrow \{\text{varieties } X \subset k^n\} \\ \{\text{prime ideals } P \text{ of } k[X_1, \dots, X_n]\} &\longleftrightarrow \{\text{irreducible varieties } X \subset k^n\} \end{aligned}$$

Therefore,

$$\text{Spec } k[X_1, \dots, X_n] = \{\text{irreducible varieties } X \subset k^n\}$$

Proposition 5.8. let $A = k[x_1, \dots, x_n]$ a fingen k -algebra (k an algebraically closed field).

Write J for the ideal of relations holding between $x_1, \dots, x_n$, so that $A = k[X_1, \dots, X_n]/J$.

Then there is a one-to-one correspondence

$$\text{Spec } A \longleftrightarrow \{\text{irreducible subvarieties } X \subset V(J)\}$$

Proof. By definition, $\text{Spec } A = \{P \mid P \subset A \text{ is prime ideal}\}$.

By Corollary 5.8:

$$\{\text{prime ideals } P \text{ of } k[X_1, \dots, X_n]\} \longleftrightarrow \{\text{irreducible varieties } X \subset k^n\}$$

About varieties:

- $I(X)$ in $R = k[X_1, \dots, X_n]$ is the ideal of the variety X
ie. the set of all polynomials that vanish on every point of X .
- $I(X)$ in $A = k[X_1, \dots, X_n]/J$, we're not looking at all possible polynomials but at the residue classes.

If P is prime in $A \implies$ it must correspond to some prime ideal $\mathfrak{P}$ in R (also $J \subset R$).

Then from Nullstellensatz (5.6), every prime ideal $\mathfrak{P}$ in R is the ideal of some irreducible variety X .

$$\implies \mathfrak{P} = I(X).$$

Since we're restricted to the ring $A = k[X_1, \dots, X_n]/J$, the ideal P are the elements of $I(X)$ viewed through the lens of the quotient

$$\implies P = \{f + J \mid f \in I(X)\} = I(X) \bmod J$$

Now, J is the set of equations defining our "universe" $V(J)$.

Since $A = R/J \implies$ we thus have $J \subseteq R$.

Also we have a correspondence between $A = R/J$ and R .

Let $\mathfrak{P}$ be the preimage of P in $R = k[X_1, \dots, X_n]$, ie.

$$\begin{aligned} R &\longrightarrow A = R/J \\ \mathfrak{P} &\longmapsto P \\ I(X) &\longmapsto I(X) \bmod J \end{aligned}$$

$$\implies \text{thus } J \subseteq \mathfrak{P}.$$

Now, $J \subset \mathfrak{P} = I(X)$;

by 5.5, the set of points where $\mathfrak{P}$ vanishes must be inside the set of points where J vanishes, so

$$V(J) \supseteq V(\mathfrak{P}) = V(I(X)) = X$$

$\implies X \subseteq V(J)$, ie. the irreducible variety X must be a subvariety of $V(J)$.

Therefore,

$$\mathfrak{P} \in \text{Spec } A \longleftrightarrow X \subseteq V(J)$$

where $X = V(\mathfrak{P})$. □

5.5 Again: Zariski, Weak Nullstellensatz, Nullstellensatz

This subsection proves again Zariski's lemma, Weak Nullstellensatz and Nullstellensatz, in a more interconnected way than the proofs described previously.

Lemma (Zariski's lemma). k a field, K a fingen k -algebra and a field.

Then K is a finite algebraic extension of k .

If k algebraically closed, then $K = k$.

Proof. K a fingen k -algebra $\iff \exists x_1, \dots, x_n \in K$ such that

$$K = k[x_1, \dots, x_n].$$

Want to show that K is algebraic over k .

Since K is a fingen k -algebra, by Noether normalization R.4.6,
 $\exists \{y_1, \dots, y_r\} \subset K$ such that

- the set $\{y_1, \dots, y_r\}$ is algebraically independent over k

- K is integral over the polynomial ring $R = k[y_1, \dots, y_r] \subset K$.

By R.4.9:

if $A \subset B$ integral extension of integral domains,

then A is a field $\iff B$ is a field.

which applied to our case:

$$B := K \text{ (a field), } A := R = k[y_1, \dots, y_r]$$

since K is integral over R and K a field $\implies$ by R.4.9 R must be a field.

Since $y_i \in K$ are algebraically independent

$\implies R = k[y_1, \dots, y_r]$ is a polynomial ring in r indeterminates.

Then, if $r \geq 1$, we have at least one variable y_1 (with $k[y_1]$);

but the only invertible elements in a polynomial ring over a field are the non-zero constants of the field k , so y_1 is not invertible in R . (alternatively: the element y_1 is not invertible in the polynomial ring, since $1/y_1$ is a rational function, not a polynomial).

Therefore, $R = k[y_1, \dots, y_r]$ can be a field iff $r = 0$, and we know that it is a field, therefore $r = 0$ and $R = k$.

By Noether normalization, at the previous step we knew that K is integral over $R (= k)$.

By 4.3.Aux, a fingen algebra that is integral over its base field is a finite vector space over that field.

Thus $[K : k] < \infty$, and K is algebraic over k .

If k is algebraically closed, every element $\alpha \in K$ is algebraic over k , and its minimal polynomial over k must have degree one. Therefore $\alpha \in k$, and thus $K = k$. $\square$

This brings us to the *Weak Nullstellensatz*:

Theorem 4.10 (Weak Nullstellensatz). k a field, K a k -algebra with

- K fingen as a k -algebra
- K a field

Then K is finite-dimensional over k , and hence algebraic over k ; that is $[K : k] < \infty$.

If k is algebraically closed, then $K = k$.

Proof. proven at previous proof. $\square$

Now, assume that k is algebraically closed, and let $m \subset k[X_1, \dots, X_n]$ be a maximal ideal. Let $A = k[X_1, \dots, X_n]$.

Then $\frac{k[X_1, \dots, X_n]}{m} = \frac{A}{m}$ is a field (by basic ideal quotient theory ($R/m = \text{Field}$)).
 Moreover, $\frac{A}{m}$ is a finite k -algebra, generated by the classes

$$X_1 + m, \dots, X_n + m.$$

Therefore, by Zariski's lemma, $\frac{A}{m}$ is algebraic over k .

And since k is algebraically closed $\implies \frac{A}{m} \cong k$.

Let

$$\pi : A \longrightarrow \frac{A}{m}$$

be the quotient map. By definition, π is the surjective homomorphism that sends each polynomial to its coset modulo m , and the kernel of this map is exactly $\ker(\pi) = m$.

By Zariski's lemma, $\frac{A}{m}$ is a finite field extension of k , and since k is algebraically closed, by Zariski's lemma, this extension must be k itself; denote this relation by

$$\theta : \frac{A}{m} \xrightarrow{\text{iso}} k$$

the k -algebra isomorphism.

Define the composite map

$$\phi = \theta \circ \pi : A \longrightarrow k.$$

which is a k -algebra homomorphism.

Thus, $\forall i \in [n]$, define

$$a_i = \phi(X_i) \in k.$$

Since ϕ is a k -algebra homomorphism, it must map any scalar $c \in k$ to itself, ie. $\phi(c) = c$. Therefore

$$\phi(X_i - a_i) = \phi(X_i) - \phi(a_i) = a_i - a_i = 0$$

Since $\phi = \theta \circ \pi$, and θ is an isomorphism (by definition injective), the only way for $\phi(f)$ to be 0 is if $\pi(f) = 0$:

$$\begin{aligned} \phi(X_i - a_i) &= \theta(\pi(X_i - a_i)) = 0 \\ \implies \pi(X_i - a_i) &= 0 \end{aligned}$$

Since $\ker(\pi) = m$, any element that maps to 0 under π must belong to m , thus

$$\pi(X_i - a_i) = 0 \implies X_i - a_i \in m$$

Alternative way: by definition of a quotient ring, and elements of $\frac{A}{m}$ are cosets of the form $f + m$. $\pi(x_i)$ sends X_i to the coset $X_i + m$, ie. $\pi(x_i) = X_i + m$. Thus, if a_i is the image of X_i , then

$$X_i \equiv a_i \pmod{m},$$

which by the definition of congruence modulo ideal, this means $X_i - a_i \in m$.

Since $X_i - a_i \in m \implies J = (X_1 - a_1, \dots, X_n - a_n) \subseteq m$,
since J is the kernel of the evaluation map at point $(a_1, \dots, a_n)$, then J is a
maximal ideal.

(Corollary 5.2) Thus $J = m$, ie. $m = (X_1 - a_1, \dots, X_n - a_n)$, with
 $a_i \in k$.

We had $k \xrightarrow{\text{iso}} \frac{A}{m} = \frac{k[X_1, \dots, X_n]}{m}$.
We have the map

$$\begin{aligned} \phi : k[X_1, \dots, X_n] &\longrightarrow \frac{k[X_1, \dots, X_n]}{m} \cong k \\ X_i &\mapsto a_i \in k \end{aligned}$$

which is the natural evaluation map $f(X_1, \dots, X_n) \mapsto f(a_1, \dots, a_n)$.

Since ϕ is a k -algebra homomorphism, then

$$\forall f \in A, \phi(f(X_1, \dots, X_n)) = f(\phi(X_1), \dots, \phi(X_n)) = f(a_1, \dots, a_n).$$

Thus, $\exists$ a one-to-one correspondence:

$$\begin{aligned} \text{points in } k^n &\longleftrightarrow m - \text{Spec } A, \text{ maximal ideals in } k[X_1, \dots, X_n] \\ (a_1, \dots, a_n) &\longleftrightarrow (X_1 - a_1, \dots, X_n - a_n) \end{aligned}$$

Now let $J \subsetneq A$ be an arbitrary proper ideal. Since every proper ideal is
contained in a maximal ideal, $\exists$ a maximal ideal m such that

$$J \subseteq m$$

We have seen that

$$m = (X_1 - a_1, \dots, X_n - a_n)$$

for some $P = (a_1, \dots, a_n) \in k^n$.

So P is a zero for every polynomial in m , ie.

$$\forall f \in m \implies f(P) = 0.$$

Since $J \subseteq m$, P is also a zero for every polynomial in J ; ie. every element
of J vanishes at P .

$\implies P \in V(J)$, and thus $V(J) \neq \emptyset$. This is what the Nullstellensatz first
part states:

Theorem 5.6 (Nullstellensatz). Let k be an algebraically closed field and let
 $J \subseteq k[X_1, \dots, X_n]$ be an ideal.

- a. if $J \subsetneq k[X_1, \dots, X_n]$ then $V(J) \neq \emptyset$
- b. $I(V(J)) = \text{rad}(J) (= \sqrt{J})$, in other words, for $f \in k[X_1, \dots, X_n]$,
- $$f(P) = 0 \ \forall P \in V(J) \iff f^N \in J \text{ for some } N \geq 1.$$

Proof. a. already proven in the previous steps.

- b. $I(V(J)) = \sqrt{J}$
 ie. vanishing ideal of a variety = radical of the ideal defining the variety
 where

$$\sqrt{J} = \{f \in A \mid f^N \in J \text{ for some } N > 0\}.$$

First, want to show that

$$\sqrt{J} \subseteq I(V(J)).$$

Let $f \in \sqrt{J}$. Then $f^N \in J$ for some $N \geq 1$.

Thus, for every $P \in V(J)$,

$$f(P)^N = f^N(P) = 0,$$

and therefore $f(P) = 0$.

Hence $f \in I(V(J))$, so

$$\sqrt{J} \subseteq I(V(J)).$$

Now, want to show that if a polynomial vanishes at all points where J vanishes, then $f \in \sqrt{J}$.

Since $k[X_1, \dots, X_n]$ is Noetherian, write

$$J = (g_1, \dots, g_m).$$

Consider the ring $k[X_1, \dots, X_n, Y]$ and the ideal J' generated by $\{g_1, \dots, g_m, 1 - Yf\}$.

Suppose there is a point $(a_1, \dots, a_n, b)$ that is a zero of J' . ie.

$$\exists (a_1, \dots, a_n, b) \in V(J').$$

Since

$$g_i(a_1, \dots, a_n) = 0,$$

our hypothesis says

$$f(a_1, \dots, a_n) = 0.$$

However, the last generator $(1 - Yf)$ requires

$$1 - bf(a_1, \dots, a_n) = 0 \implies 1 - b \cdot 0 = 0 \implies 1 = 0,$$

a contradiction.

Therefore, $V(J') = \emptyset$.

By part a., applied to $k[X_1, \dots, X_n, Y]$,
if $V(J') = \emptyset$ then $J' = k[X_1, \dots, X_n, Y]$, so $1 \in J'$.

Every element in an ideal is a linear combination of its generators:
 J' is generated by $\{g_1, \dots, g_m, 1 - Yf\}$

$$\implies \forall j \in J', \quad j = \left(\sum (\text{polynomial}) g_i \right) + (\text{polynomial}) \cdot (1 - Yf).$$

which, since $1 \in J'$,

$$1 = \left(\sum_{i=1}^m p_i(X, Y) g_i(X) \right) + q(X, Y) \cdot (1 - Yf(X)).$$

Now work in the localization

$$A_f = A[f^{-1}],$$

where f is invertible, and apply the homomorphism

$$A[Y] \longrightarrow A_f, \quad Y \longmapsto f^{-1}.$$

Then

$$1 = \left(\sum_{i=1}^m p_i(X, f^{-1}) g_i(X) \right) + q(X, f^{-1}) \cdot \underbrace{(1 - f^{-1}f(X))}_0,$$

thus

$$1 = \sum_{i=1}^m p_i(X, f^{-1}) g_i(X).$$

Since there are finitely many polynomials p_i , $\exists N \geq 0$ such that

$$f^N p_i(X, f^{-1}) \in A \quad \forall i.$$

Multiplying by f^N ,

$$f^N = \sum_{i=1}^m A_i(X) g_i(X),$$

where

$$A_i(X) = f^N p_i(X, f^{-1}) \in A.$$

Thus f^N is a linear combination of g_i .
Therefore $f^N \in J$, so $f \in \sqrt{J}$.

Hence

$$I(V(J)) \subseteq \sqrt{J}.$$

Combining both inclusions,

$$I(V(J)) = \sqrt{J}.$$

□

6 Rings of fractions $S^{-1}A$ and localization

6.1 Rings of fractions $S^{-1}A$

Definition 6.1 (ring of fractions). let A a ring, $S \subset A$ a multiplicative set ($1 \in S$, and $st \in S \forall s, t \in S$).

Introduce the following relation $\sim$ on $A \times S$:

$$(a, s) \sim (b, t) \iff \exists y \in S \text{ such that } u(at - bs) = 0$$

(write a/s for the equivalence class of (a, s)).

Then, the *ring of fractions of A with respect to S* is

$$S^{-1}A = (A \times S) / \sim$$

with ring op'ns defined by the usual arithmetic op'ns on fractions:

$$\frac{a}{s} \pm \frac{b}{t} = \frac{(at \pm bs)}{st} \quad \text{and} \quad \frac{a}{s} \cdot \frac{b}{t} = \frac{ab}{st}$$

Proposition 6.1. i. $\sim$ is an equivalence relation

ii. the ring op'ns are well defined, and $S^{-1}A$ is a ring

iii.

$$\begin{aligned} \psi : A &\longrightarrow S^{-1}A \\ a &\longmapsto a/1 \end{aligned}$$

is a ring homomorphism.

Example . TODO

Lemma 6.2. For $f \in A$, write $S = \{1, f, f^2, \dots\}$, and $A_f \cong S^{-1}A$.

Then

$$A_f \cong \frac{A[X]}{(Xf - 1)}$$

Proof. Define the homomorphism

$$\begin{aligned}\psi : A[X] &\longrightarrow A_f \\ a &\longmapsto a/1 \\ X &\longmapsto 1/f\end{aligned}$$

By the 1st isomorphism theorem:

$$\begin{array}{ccc} A[X] & \xrightarrow{\psi} & A_f \\ \searrow \phi & & \nearrow \eta \\ & A[X]/\ker(\psi) & \end{array}$$

Thus we want to prove that $\ker(\psi) = (Xf - 1)$, so that $\frac{A[X]}{\ker(\psi)} = \frac{A[X]}{(Xf-1)}$, and the lemma is proven.

First, observe that $\psi(Xf - 1) = \psi(X)\psi(f) - \psi(1) = \frac{1}{f} \frac{f}{1} - 1 = 1 - 1 = 0$, so $Xf - 1 \in \ker(\psi)$, ie. $(Xf - 1) \subseteq \ker(\psi)$.

Now, we want to prove that $\ker(\psi) \subseteq (Xf - 1)$.

Take $h \in \ker(\psi)$, will prove that $h \in (Xf - 1) \forall h \in \ker(\psi)$, and thus $\ker(\psi) \subseteq (Xf - 1)$.

Want to prove that $h(X)$ is a multiple of $(Xf - 1)$.

Let

$$h(X) = a_n X^n + a_{n-1} X^{n-1} + \dots + a_1 X + a_0$$

multiply $h(X)$ by f^n :

$$f^n \cdot h(X) = a_n(f^n X^n) + a_{n-1}f(f^{n-1} X^{n-1}) + a_{n-2}f^2(f^{n-2} X^{n-2}) \dots$$

Note that since $\forall i \geq 1, f^i X^i = (Xf - 1) \cdot (f^{i-1} X^{i-1} + f^{i-2} X^{i-2} + \dots + 1)$, then $f^i X^i \equiv 1 \pmod{Xf - 1}$.

So,

$$f^n \cdot h(X) = \underbrace{a_n(1) + a_{n-1}f(1) + a_{n-2}f^2(1) + \dots + a_0 f^n}_C \pmod{Xf - 1}$$

$C \text{ (constant)}$

$$\implies f^n \cdot h(X) = C \pmod{Xf - 1}$$

$$\iff f^n \cdot h(X) = Q(X) \cdot (Xf - 1) + C$$

Want to remove C , but it is non-zero. Note that in A_f (ring of fractions), $\frac{a}{f^n} = 0$ iff $\exists k$ such that $f^k \cdot a = 0$ in A .

So, multiply both sides by f^k :

$$f^k \cdot f^n \cdot h(X) = f^k \cdot (Q(X) \cdot (Xf - 1) + C)$$

$$\underbrace{f^k f^n}_{f^{nk}} \cdot h(X) = \underbrace{f^k Q(X)}_{Q'(X)} \cdot (Xf - 1) + \underbrace{f^k C}_0$$

$$\begin{aligned} \implies f^{n+k} \cdot h(X) &= Q'(X) \cdot (Xf - 1) \\ \iff f^{n+k} \cdot h(X) &\equiv 0 \pmod{Xf - 1} \end{aligned}$$

multiply it by X^{n+k} :

$$\begin{aligned} X^{n+k} \cdot f^{n+k} \cdot h(X) &\equiv X^{n+k} \cdot 0 \pmod{Xf - 1} \\ (Xf)^{n+k} \cdot h(X) &\equiv 0 \pmod{Xf - 1} \end{aligned}$$

Now, since we had $Xf \equiv 1$ in $\frac{A[X]}{(Xf-1)}$,

$$\begin{aligned} (1)^{n+k} \cdot h(X) &\equiv 0 \pmod{Xf - 1} \\ \implies h(X) &\equiv 0 \pmod{Xf - 1} \end{aligned}$$

By definition this is saying $h(X) \in (Xf - 1) \forall k \in \ker(\psi)$.

Thus $\ker(\psi) \subseteq (Xf - 1)$.

Initially we saw that $(Xf - 1) \subseteq \ker(\psi)$. Therefore $\ker(\psi) = (Xf - 1)$.

Hence,

$$A_f \cong \frac{A[X]}{(Xf - 1)}$$

□

Given a ring homomorphism $\psi : A \longrightarrow B$, there is a correspondence

$$e : \{\text{ideals of } A\} \longrightarrow \{\text{ideals of } B\}$$

given by $e(I) = \psi(I)B = IB$ (called *extension*),

and

$$r : \{\text{ideals of } B\} \longrightarrow \{\text{ideals of } A\}$$

given by $r(J) = \psi^{-1}J$ (called *restriction*, written $A \cap J$).

Set $B = S^{-1}A$. Then $S^{-1}I = e(I) = \psi(I)B$.

Proposition 6.3. a. $\forall$ ideal J of $S^{-1}A$, $e(r(J)) = J$

b. $\forall$ ideal I of A ,

$$r(e(I)) = \{a \in A \mid as \in I \text{ for some } s \in S\}$$

c. if P prime and $P \cap S = \emptyset$,
then $e(P) = S^{-1}P$ is a prime ideal of $S^{-1}A$.

Proof.

□

Corollary 6.3. for an ideal I of A , the necessary and sufficient condition for $r(e(I)) = I$ is

$$as \in I \implies a \in I \quad \forall s \in S$$

6.2 Localization

If P prime ideal, then $S = A \setminus P$ is a multiplicative set. Define the set $A_P = S^{-1}A$.

Proposition 6.4. $a/s \in A_P$ is a unit of $A_P \iff a \notin P$.

Therefore A_P is a *local ring*, with maximal ideal $e(P) = PA_P$.

The local ring (A_P, PA_P) is called the *localization* of A at P .

Proof. $\Leftarrow$ (if $a \notin P \implies a/s \in A_P$ is a unit)

if $a \notin P$, then by definition of S , $a \in S$.

In the localization A_P , every element of S is invertible.

The inverse of $\frac{a}{s}$ is $\frac{s}{a}$

$$\longrightarrow \frac{a}{s} \cdot \frac{s}{a} = 1, \text{ thus } a/s \in A_P \text{ is a unit.}$$

$\implies$ (if a/s unit $\implies a \notin P$)

Suppose a/s a unit; then $\exists \frac{b}{t} \in A_P$ such that $\frac{a}{s} \cdot \frac{b}{t} = 1$.

By definition of equality in localization, $\frac{ab}{st} = 1$ means $\exists u \in S$ such that

$$u(ab \cdot 1 - st \cdot 1) = 0 \implies uab = ust \quad (8)$$

with $u, s, t \in S$.

Since $S = A \setminus P$ and P is a prime ideal,

the products of elements outside P must also be outside of P .

$$u, s, t \notin P \implies ust \notin P$$

At (8), we know that $uab = ust \notin P$, thus $uab \notin P$.

If $uab \notin P$, then $a \notin P$.

Next we will prove that A_P is a local ring.

A ring is local if it has exactly one maximal ideal (1.13); so we show that the set of non-units forms an ideal.

a/s is not a unit iff $a \in P$. Let $m = \{a/s | a \in P, s \notin P\} = PA_P$. Want to show that PA_P is the unique maximal ideal:

- it's an ideal: let $\frac{a}{s}, \frac{b}{t} \in PA_P$ with $a, b \in P$.
Then $\frac{a}{s} + \frac{b}{t} \in PA_P$ with numerator in P .
If multiply a fraction in PA_P by any fraction in A_P , the numerator stays in P , thus PA_P is an ideal.
- every element outside of PA_P is a unit (proven at the beginning of this proposition's proof ($\Leftarrow$)).

- maximality: every ideal strictly larger than PA_P must contain a unit.
 If an ideal contains a unit, it must be the entire ring A_P .
 $\implies$ therefore, PA_P is the unique maximal ideal.

□

In other words:

Localization: formal way to 'force' certain elements to have inverses.

It's the smallest ring that contains A and makes all the elements of S invertible.

There is a natural map $\psi : A \longrightarrow S^{-1}A$, by $f(a) = \frac{a}{1}$, $f(s) = \frac{1}{s}$ for $s \in S$.

6.3 Localization commutes with taking quotients

Let A ring, S multiplicative set, I ideal.

Write T for the image of S in A/I .

Then $S^{-1}I = I \cdot S^{-1}A$ is an ideal of $S^{-1}A$, can take the quotient ring $S^{-1}A/S^{-1}I$.

$\longleftrightarrow$ can take the quotient A/I and then localize to get $T^{-1}(A/I)$.

Corollary 6.7.

$$T^{-1}(A/I) \cong \frac{S^{-1}A}{S^{-1}I}$$

In particular, for P prime ideal,

$$\frac{A_P}{PA_P} = k(P) = \text{Frac}(\underbrace{A/P}_{\text{integral domain}})$$

From 6.4, A_P : local ring, PA_P : unique maximal ideal. $k(P)$ and $\text{Frac}(A/P)$ are field of fractions.

Proof. the quotient ring A/I can be viewed as an A -module and

$$\underbrace{T^{-1}(A/I)}_{\text{ring of fractions}} \cong \underbrace{\frac{S^{-1}A}{S^{-1}I}}_{\text{module of fractions}}$$

The ??i, gives an isomorphism of modules

$$T^{-1}(A/I) = S^{-1}(A/I) \cong \frac{S^{-1}A}{S^{-1}I}$$

it's easy to see that this is a ring homomorphism.

□

7 Tensor Product of Modules

7.1 Constructing the Tensor Product

We use vector spaces but the same applies for modules.

Let Z be the vector space with basis $\{e_{v,w}\} \forall (v,w) \in V \times W$.

Want to introduce the relation $a = b \in Z$. Consider the element $a - b \in Z$; take the span $Span\{a - b\}$, that is, all of multiples of $a - b$.

Consider the quotient space $\frac{Z}{Span\{a-b\}}$. Anything in the span of $a - b$ is equal to zero.

$\implies$ This is how we use relations to get vector spaces out of Z .

Direct Sum, $V \oplus W$.

Want

- i. $(v_1, w_1) + (v_2, w_2) = (v_1 + v_2, w_1 + w_2)$
- ii. $\lambda(v, w) = (\lambda v, \lambda w)$

Take the quotient

$$Z / Span \left\{ \begin{array}{l} (v_1, w_1) + (v_2, w_2) - (v_1 + v_2, w_1 + w_2) \\ \lambda(v, w) - (\lambda v, \lambda w) \end{array} \right\}$$

Tensor Product, $V \otimes W$ relations:

- i. distributivity law:

$$a(b + c) = ab + ac$$

$$(a + b)c = ac + bc$$

- ii. scalar multiplication:

$$\lambda(a \cdot b) = (\lambda a)b = a(\lambda b)$$

$\implies$

$$a(b + c) = ab + ac$$

$$(a + b)c = ac + bc$$

$$\lambda(a \cdot b) = (\lambda a)b$$

$$\lambda(a \cdot b) = a(\lambda b)$$

then we can define the tensor product as

$$V \otimes W = Z / Span \left\{ \begin{array}{l} a \otimes (b+c) - a \otimes b - a \otimes c \\ (a+b) \otimes c - a \otimes c - b \otimes c \\ \lambda(a \otimes b) - (\lambda a) \otimes b \\ \lambda(a \otimes b) - a \otimes (\lambda b) \end{array} \right\}$$

7.2 Universal Property of Tensor Products

Proposition (Universal Property of Tensor Products). For every bilinear map $g : V \times W \rightarrow U$, $\exists!$ linear map $f : T \rightarrow U$ such that $f \circ \tau = g$.

$$\begin{array}{ccc} V \times W & \xrightarrow{\tau} & T = V \otimes W \\ \downarrow g & \nearrow f & \\ U & & \end{array}$$

Proof. i. uniqueness:

suppose we have two different T.P., (T, τ) and (T', τ') . Want to prove $T \stackrel{iso}{\cong} T'$.

On the right is the same relation as on the left, but switching T by T' :

$$\begin{array}{ccc} V \times W & \xrightarrow{\tau} & T \\ \downarrow \tau' & \nearrow f & \\ T' & & \end{array}$$

by the U.P., $f \circ \tau = \tau'$.

$$\begin{array}{ccc} V \times W & \xrightarrow{\tau} & T \\ \downarrow \tau' & \nearrow f & \\ T' & & \end{array}$$

by the U.P., $f \circ \tau = \tau'$.

$$\text{thus, } f \circ f' \circ \tau' = \tau'$$

Want to show that f has an inverse, so that it is an isomorphism $T \stackrel{iso}{\cong} T'$.
we just saw that

$$f \circ f' \circ \tau' = \tau'$$

also we can get it the other way

$$f' \circ f \circ \tau = \tau$$

we can see that $f \circ f'$ seems to be the identity, need more info to prove it.

What if we apply the U.P. to the same module on both sides?

$$\begin{array}{ccc} V \times W & \xrightarrow{\tau} & T \\ \downarrow \tau & \nearrow ? = id_T & \\ T & & \end{array}$$

thus, $id_T \circ \tau = \tau$, hence

$$\begin{array}{ccc} V \times W & \xrightarrow{\tau} & T \\ \tau \downarrow & \nearrow id_T, f' \circ f & \\ T & & \end{array}$$

and by uniqueness of the U.P., we can not have two different functions (id_T and $f' \circ f$) and must be that $f' \circ f = id_T$.

Similarly for $f \circ f' = id_{T'}$.

Therefore, f' is the inverse of f , and $T \cong^{iso} T'$, which means that the tensor prodduct is unique.

This is stated in the Atiyah and MacDonald book ([1]) for multiple nodes and multilinear maps:

Proposition 2.12. Let $M_1, \dots, M_r$ be A -modules. Then $\exists (T, g)$ with T an A -module, and g a multilinear map $g : M_1 \times \dots \times M_r \longrightarrow T$, with the following property:

Given any A -module P and any A -multilinear map $f : M_1 \times \dots \times M_r \longrightarrow P$, $\exists!$ an A -homomorphism $f' : T \longrightarrow P$ such that $f' \circ g = f$.

Moreover, if (T, g) and (T', g') are two pairs with this property, then $\exists! j : T \xrightarrow{iso} T'$ such that $j \circ g = g'$.

7.3 Usefulness of Tensor Product's Universal Property

Anytime we define a function on the T.P. we need to know that it is well defined and linear. This is hard in general because of all the relations of the elements of the T.P..

But using the U.P. of T.P. we can remove all of the extra work and just look at a bilinear map on the cartesian product, which is a much easier object to work with.

Using the U.P. to show that a function on the T.P. is well defined and linear:

- i. define a map ϕ on the Cartesian product

$$(x, y) \in M \times N$$

- ii. prove bilinearity of ϕ :
show that ϕ is A -linear in each variable separately.

ie. check

$$\begin{aligned}\phi(x + x', y) &= \phi(x, y) + \phi(x', y) \\ \phi(x, y + y') &= \phi(x, y) + \phi(x, y') \\ \phi(ax, y) &= a\phi(x, y) = \phi(x, ay)\end{aligned}$$

iii. invoke the Universal Property.

Once bilinearity is proven on ϕ , the U.P. guarantees the existence of a unique map f on the tensor product $M \otimes N$

iv. automatic well-definedness and linearity: f is linear by definition.

And since it is a single map originating from the bilinear map ϕ on the product, it is automatically well-defined on the tensor product (ie. automatically "respects" the relations that define tensors).

□

8 Exercises

For the exercises, I follow the assignments listed at [3].

The exercises that start with **R** are the ones from the book [2], and the ones starting with **AM** are the ones from the book [1].

8.1 Exercises Chapter 1

Exercise R.1.1. Ring A and ideals I, J such that $I \cup J$ is not an ideal. What's the smallest ideal containing I and J ?

Proof. Take ring $A = \mathbb{Z}$. Set $I = 2\mathbb{Z}$, $J = 3\mathbb{Z}$.

I, J are ideals of $A (= \mathbb{Z})$. And $I \cup J = 2\mathbb{Z} \cup 3\mathbb{Z}$.

Observe that for $2 \in I, 3 \in J \implies 2, 3 \in I \cup J$, but $2 + 3 = 5 \notin I \cup J$.

Thus $I \cup J$ is not closed under addition; thus is not an ideal.

Smallest ideal of $\mathbb{Z} (= A)$ containing I and J is their sum:

$$I + J = \{a + b \mid a \in I, b \in J\}$$

$\gcd(2, 3) = 1$, so $I + J = \mathbb{Z}$.

Therefore, smallest ideal containing I and J is the whole ring $\mathbb{Z}$. $\square$

Exercise R.1.5. let $\psi : A \longrightarrow B$ a ring homomorphism. Prove that ψ^{-1} takes prime ideals of B to prime ideals of A .

In particular if $A \subset B$ and P a prime ideal of B , then $A \cap P$ is a prime ideal of A .

Proof. (Recall: prime ideal is if $a, b \in R$ and $a \cdot b \in P$ (with $R \neq P$), implies $a \in P$ or $b \in P$).

Let

$$\psi^{-1}(P) = \{a \in A \mid \psi(a) \in P\} = A \cap P$$

The claim is that $\psi^{-1}(P)$ is prime ideal of A .

i. show that $\psi^{-1}(P)$ is an ideal of A :

$0_A \in \psi^{-1}(P)$, since $\psi(0_A) = 0_B \in P$ (since every ideal contains 0).

If $a, b \in \psi^{-1}(P)$, then $\psi(a), \psi(b) \in P$, so

$$\psi(a - b) = \psi(a) - \psi(b) \in P$$

hence $a - b \in \psi^{-1}(P)$.

If $a \in \psi^{-1}(P)$ and $r \in A$, then $\psi(ra) = \psi(r)\psi(a) \in P$, since P is an ideal.

Thus $ra \in \psi^{-1}(P)$.

$\implies$ so ψ^{-1} is an ideal of A .

ii. show that $\psi^{-1}(P)$ is prime:

$\psi^{-1}(P) \neq A$, since if $\psi^{-1}(P) = A$, then $1_A \in \psi^{-1}(P)$, so $\psi(1_A) = 1_B \in P$, which would mean that $P = B$, a contradiction since P is prime ideal of B .

Take $a, b \in A$ with $ab \in \psi^{-1}(P)$; then $\psi(ab) \in P$, and since ψ is a ring homomorphism, $\psi(ab) = \psi(a)\psi(b)$.

Since P prime ideal, then $\psi(a)\psi(b) \in P$ implies either $\psi(a) \in P$ or $\psi(b) \in P$. Thus $a \in \psi^{-1}(P)$ or $b \in \psi^{-1}(P)$.

Hence $\psi^{-1}(P)$ ($= A \cap P$) is a prime ideal of A .

□

Exercise R.1.6. prove or give a counter example:

- the intersection of two prime ideals is prime
- the ideal $P_1 + P_2$ generated by 2 prime ideals P_1, P_2 is prime
- if $\psi : A \longrightarrow B$ ring homomorphism, then ψ^{-1} takes maximal ideals of B to maximal ideals of A
- the map ψ^{-1} of Proposition 1.2 takes maximal ideals of A/I to maximal ideals of A

Proof. a. let $I = 2\mathbb{Z} = (2)$, $J = 3\mathbb{Z} = (3)$ be ideals of $\mathbb{Z}$, both prime.

Then $I \cap J = (2) \cap (3) = (6)$.

The ideal (6) is not prime in $\mathbb{Z}$, since $2 \cdot 3 \in (6)$, but $2 \notin (6)$ and $3 \notin (6)$.

Thus the intersection of two primes can not be prime.

- $P_1 = (2)$, $P_2 = (3)$, both prime.

Then,

$$P_1 + P_2 = (2) + (3) = \{a + b | a \in P_1, b \in P_2\}$$

$\longrightarrow$ in a principal ideal domain (like $\mathbb{Z}$), the sum of two principal ideals is again principal, and given by $(m) + (n) = (gcd(m, n))$.

(recall: principal= generated by a single element)

So, $P_1 + P_2 = (2) + (3) = (gcd(2, 3)) = (1) = \mathbb{Z}$.

The whole ring is not a prime ideal (by the definition of the prime ideal), so $P_1 + P_2$ is not a prime ideal.

Henceforth, the sum of two prime ideals is not necessarily prime.

- let $A = \mathbb{Z}$, $B = \mathbb{Q}$, $\psi : A \longrightarrow B$.

Since $\mathbb{Q}$ is a field, its only maximal ideal is (0) .

Then

$$\psi^{-1}((0)) = (0) \subset \mathbb{Z}$$

$$\text{ie. } \psi^{-1}(m_B) = (m_B) \subset A$$

But (0) is not maximal in $\mathbb{Z}$, because $\mathbb{Z}/(0) \cong \mathbb{Z}$ is not a field.

Thus the preimages of maximal ideals under arbitrary ring homomorphisms need not be maximal.

d. $\psi : A \longrightarrow A/I$ quotient homomorphism, $I \subseteq A$ an ideal.

Let M a maximal ideal of A/I , then $\frac{(A/I)}{M}$ is a field (Proposition 1.3).

By the isomorphism theorems,

$$\frac{(A/I)}{M} \cong \frac{A}{\psi^{-1}(M)}$$

Since $\frac{(A/I)}{M}$ is a field, the quotient $\frac{A}{\psi^{-1}(M)}$ is a field, so $\psi^{-1}(M)$ is a maximal ideal of A .

$\implies$ under ψ , preimages of maximal ideals are maximal.

□

Exercise R.1.12.a. if I, J ideals and P prime ideal, prove that

$$IJ \subseteq P \iff I \cap J \subseteq P \iff I \text{ or } J \subseteq P$$

Proof. assume $I \subseteq P$ (for $J \subseteq P$ will be the same, symmetric), take $x \in IJ$, then

$$x = \sum_{k=1}^n a_k b_k$$

with $a_k \in I$, $b_k \in J$.

Each $a_k \in I \subseteq P$. Since P an ideal,

$$\sum_{k=1}^n a_k b_k \in P$$

thus $x \in P$, hence $IJ \subseteq P$.

So $I \subseteq P$ or $J \subseteq P \implies IJ \subseteq P$.

Conversely,

assume P prime and $IJ \subseteq P$.

Suppose by contradiction that $I \not\subseteq P$ and $J \not\subseteq P$.

- since $I \not\subseteq P$, $\exists a \in I$ with $a \notin P$

- since $J \not\subseteq P$, $\exists b \in J$ with $b \notin P$

Since $a \in I$, $b \in J$, $ab \in IJ \subseteq P$, but P is prime, so $ab \in P$ implies that $a \in P$ or $b \in P$. This contradicts a, b being taken outside of P .

Thus $I \not\subseteq P$ and $J \not\subseteq P$ are false.

So both directions are proven, hence

$$IJ \subseteq P \implies I \subseteq P \text{ or } J \subseteq P$$

□

Exercise R.1.18. Use Zorn's lemma to prove that any prime ideal P contains a minimal prime ideal.

Proof. Let P prime ideal of R .

$$S = \{Q \subseteq R \mid Q \text{ a prime ideal AND } Q \subseteq P\}$$

Goal: show that S has a minimal element, the minimal ideal contained in P .

$P \subset S$, so S is nonempty.

Let $C \subseteq S$ be a chain (= totally ordered subset) with respect to inclusion.

Define

$$Q_C = \bigcap_{Q \in C} Q$$

Clearly $Q_C \subseteq P$, since each $Q \in C$ is $Q \subseteq P$.

Since C is ordered by inclusion, it is a decreasing chain of prime ideals.

Intersection of a decreasing chain of prime ideals is again a prime ideal:

- if $ab \in Q_C$, then $ab \in Q \forall Q \in C$
- since Q prime, $\forall Q \in C$ either $a \in Q$ or $b \in Q$

If there were some $Q_1, Q_2 \in C$ with $a \in Q_1$ and $b \notin Q_2$, then by total ordering, either $Q_1 \subseteq Q_2$ or $Q_2 \subseteq Q_1$.

In either case: contradiction, since the smaller one would have to contain the element that was assumed to be excluded.

Thus $\forall Q \in C$ the same element a, b must lie in all Q . $\implies$ lies in the intersection of them, Q_C .

Henceforth, Q_C is a prime ideal and lies in S , and its a lower bound of C in S .

Now, S is nonempty, and every chain in S has a lower bound in S (its intersection).

Therefore, S has a minimal element P_{min} .

By construction, P_{min} is a prime ideal $P_{min} \subseteq P$, and by minimality there are no strictly smaller prime ideals inside P .

So P_{min} is a minimal prime ideal, contained in P . □

Exercise R.1.10.

Proof. □

Exercise R.1.11.

Proof. □

Exercise R.1.4.

Proof. □

8.2 Exercises Chapter 2

Exercise R.2.9. $0 \longrightarrow L \xrightarrow{\alpha} M \xrightarrow{\beta} N \longrightarrow 0$ is a s.e.s. of A -modules. Prove that if N, L are finite over A , then M is finite over A .

Proof. Denote the generators of L and N respectively as

$$\begin{aligned} \{l_1, \dots, l_k\} &\subseteq L \\ \{n_1, \dots, n_p\} &\subseteq N \end{aligned}$$

By s.e.s. definition,

- α is injective (one-to-one), so

$$\forall l_i \in L, \exists x_i \in M \text{ s.th. } \alpha(l_i) = x_i$$

- β is surjective (onto), so

$$\forall n_j \in N, \exists y_j \in M \text{ s.th. } \beta(y_j) = n_j$$

We will show that $\{x_1, \dots, x_k, y_1, \dots, y_p\}$ generate M , and thus M is finite:
Let $m \in M$, then $\beta(m) \in N$, and

$$\beta(m) = \sum_{j=1}^p a_j n_j \quad \text{with } a_j \in A$$

Take $m' \in M$, with $m' = \sum a_j y_j$, then

$$\beta(m') = \sum a_j \beta(y_j) = \sum a_j n_j = \beta(m)$$

Then, since $\beta(m) = \beta(m') \implies \beta(m - m') = 0$, thus

$$(m - m') \in \ker(\beta)$$

By *exactness* property, since $\alpha : L \longrightarrow \ker(\beta)$, we have $\ker(\beta) = \text{im}(\alpha)$.

Therefore, $\exists l \in L$ such that $\alpha(l) = m - m'$.

Since $\{l_i\}_k$ generate L ,

$$l = \sum_{i=1}^k b_i l_i$$

thus

$$m - m' = \alpha(l) = \alpha\left(\underbrace{\sum_{i=1}^k b_i l_i}_l\right) = \sum_{i=1}^k b_i \underbrace{\alpha(l_i)}_{x_i} = \sum_{i=1}^k b_i x_i$$

Rearrange,

$$m = m' + \sum_{i=1}^k b_i x_i = \sum_{j=1}^p a_j y_j + \sum_{i=1}^k b_i x_i \quad \forall m \in M$$

So, L provides k generators for the kernel part of M , N provides p "lifts" for the quotient part of M ; thus M is generated by $k + p$ elements.
Thus M is finitely generated over A . □

8.3 Exercises Chapter 3

Exercise R.3.2. K a field, $A \supset K$ a ring which is finite dimensional as a K -vector space. Prove that A is Noetherian and Artinian.

Proof. $\dim(A) = n < \infty$, so every ideal $\mathfrak{a}$ of A is a K -subspace of A , because if $x \in \mathfrak{a}$ and $c \in K$, then $c \cdot x \in \mathfrak{a}$.

1. Noetherian:

let $I_1 \subseteq I_2 \subseteq \dots$ be an ascending chain of ideals in A .

Since each I_i is a subspace, we have

$$\dim_K(I_1) \leq \dim_K(I_2) \leq \dots \leq n$$

where at some $i = m$ we have $\dim_K(I_m) = \dim_K(I_{m+1})$; then since $I_m \subseteq I_{m+1}$, we have $I_m = I_{m+1}$. So A is Noetherian.

2. Artinian:

Similarly, if $I_1 \supseteq I_2 \supseteq \dots$ a descending chain of ideals in A .

then

$$n \geq \dim_K(I_1) \geq \dim_K(I_2) \geq \dots \geq 0$$

where at some $i = m$ we have $\dim_K(I_m) = \dim_K(I_{m+1})$; then since $I_m \supseteq I_{m+1}$, we have $I_m = I_{m+1}$. So A is Artinian.

□

Exercise R.3.5. Let $0 \longrightarrow L \xrightarrow{\alpha} M \xrightarrow{\beta} N \longrightarrow 0$ an exact sequence. Let $M_1, M_2 \subseteq M$ be submodules of M .

Prove if the following holds or not:

$$\beta(M_1) = \beta(M_2) \text{ and } \alpha^{-1}(M_1) = \alpha^{-1}(M_2) \implies M_1 = M_2$$

Proof. Counterexample showing that it does not hold:

Let K a field, $M = K \oplus K$, $L = K$, $N = K$.

Set, for $l \in L$, $(m_1, m_2) \in M$,

$$\alpha : l \mapsto (l, 0)$$

$$\beta : (m_1, m_2) \mapsto m_2$$

So we have

$$0 \longrightarrow K \xrightarrow{\alpha} K^2 \xrightarrow{\beta} K \longrightarrow 0$$

Then,

$$M_1 = \{(x, x) \mid x \in K\} \sim (\text{diagonal line})$$

$$M_2 = \{(0, x) \mid x \in K\} \sim (\text{y-axis})$$

(Geometric interpretation: M_1 , M_2 are the *diagonal line* and *y-axis* respectively; and α , β capture information about the *vertical* components (x-axis,

y-axis respectively), but not about the *diagonal* way a submodule is embedded in M).

Then,

$$\begin{aligned}\beta(M_1) &= \{x \mid x \in K\} = K \\ \beta(M_2) &= \{x \mid x \in K\} = K\end{aligned}$$

thus, $\beta(M_1) = \beta(M_2)$.

For M_1 , $(l, 0) \in M$ iff $l = 0$, thus $\alpha^{-1}(M_1) = \{0\}$,
for M_2 , $(l, 0) \in M$ iff $l = 0$, thus $\alpha^{-1}(M_2) = \{0\}$,
thus $\alpha^{-1}(M_1) = \alpha^{-1}(M_2)$.

So we've seen that

$$\begin{aligned}\beta(M_1) &= \beta(M_2) \\ \alpha^{-1}(M_1) &= \alpha^{-1}(M_2)\end{aligned}$$

while having $M_1 \neq M_2$. □

Exercise R.3.3. Let A a ring, $I_1, \dots, I_k$ ideals such that each A/I_i is a Noetherian ring. Prove that $\bigoplus A/I_i$ is a Noetherian A -module, and deduce that if $\bigcap I_i = 0$ then A is also Noetherian.

Proof. i. by Corollary R.3.5 (i), if M_i Noetherian modules, then $\bigoplus M_i$ is Noetherian. $\implies$ thus $\bigoplus A/I_i$ is Noetherian.

ii. Take the canonical homomorphism

$$\phi : A \longrightarrow \bigoplus_{i=1}^n A/I_i$$

by $\phi(a) = (a + I_1, a + I_2, \dots, a + I_n)$.

ϕ is injective: $\ker(\phi) = \{a \in A \mid a \in I_i \forall i\}$.

Since we're given $\bigcap I_i = 0$, then $\ker(\phi) = \bigcap I_i$, and ϕ is injective.

Thus, ϕ is the isomorphism $A \cong \text{im}(\phi)$, where $\text{im}(\phi)$ is an A -submodule of $\bigoplus A/I_i$.

We know that any submodule of a Noetherian module is Noetherian, thus, since

- A/I_i is Noetherian by hypothesis of the exercise
- $A \cong \text{im}(\phi)$
- $\text{im}(\phi)$ is an A -submodule of $\bigoplus A/I_i$

then, A is Noetherian. □

Exercise R.3.4. Prove that if A is a Noetherian ring and M a finite A -module, then there exists an exact sequence $A^q \xrightarrow{\alpha} A^p \xrightarrow{\beta} M \rightarrow 0$. That is, M has a presentation as an A -module in terms of finitely many generators and relations.

Proof. since M fingen $\implies$ generators $\{m_1, \dots, m_2\} \subseteq M$ span M .

Let β be a surjective A -linear map, which forms a free A -module of rank p onto M :

$$\begin{aligned} \beta : A^p &\longrightarrow M \\ (a_1, \dots, a_p) &\longmapsto \sum_{i=1}^p a_i m_i \end{aligned}$$

Let $K = \ker(\beta)$. By the 1st Isomorphism Theorem,

$$M \cong A^p / K$$

Since A is a Noetherian ring, then every free A -module of finite rank (eg. A^p) is a Noetherian module.

Every submodule of a Noetherian module is fingen.

$\implies$ since $K \subseteq A^p$, $\implies K (= \ker(\beta))$ is fingen.

Since K fingen, let $\{k_1, \dots, k_q\}$ be generators of K .

Define $\psi : A^q \longrightarrow K$.

Compose it with the inclusion map $i : K \longrightarrow A^p$,

$$\alpha = i \circ \psi : A^q \longrightarrow A^p$$

So we have the whole sequence $A^q \xrightarrow{\alpha} A^p \xrightarrow{\beta} M \rightarrow 0$, where

- β is surjective
- $\text{im}(\alpha) = \ker(\beta)$

so that it is a exact sequence, thus, M has a finite presentation. $\square$

8.4 Exercises Chapter 4

Exercise R.4.1.a. $k[X^2] \subset k[X]$ is a finite extension, hence integral. Find the integral dependence relation for any $f \in k[X]$.

Proof. $\forall f(X) \in k[X]$ can be uniquely decomposed into its even and odd parts:

$$f(X) = p(X^2) + X \cdot q(X^2)$$

with $p(X^2), q(X^2) \in k[X^2]$, and

$p(X^2)$: sum of all terms with even exponents

$q(X^2)$: sum of all terms with odd exponents, and then factoring out X .

(Observation: this is used in FRI cryptographic protocol

https://github.com/arnaucube/math/blob/master/notes_fri_stir.pdf)

Rearrange it

$$\begin{aligned}
f(X) - p(X^2) &= X \cdot q(X^2), \quad \text{square:} \\
(f(X) - p(X^2))^2 &= X^2 \cdot q(X^2)^2 \\
f(X)^2 - 2p(X^2)f(X) + p(X^2)^2 &= X^2 \cdot q(X^2)^2 \\
f(X)^2 - \underbrace{[2p(X^2)]}_{a_1} f(X) + \underbrace{[p(X^2)^2 - X^2 \cdot q(X^2)^2]}_{a_0} &= 0
\end{aligned}$$

Denote the last polynomial as $P(T) \in k[X^2]$, where $f(X)$ is a root of $P(T)$.

The integral dependence relation for any $f \in k[X]$ is given by the monic polynomial from R.4.1.ii, in this case $T^2 + a_1T + a_0 = 0$ with $a_i \in k[X^2]$.

We have that

$$\begin{aligned}
a_1 &= -2p(X^2) \\
a_0 &= p(X^2)^2 - X^2q(X^2)^2
\end{aligned}$$

So for example, for $f(X) = X^3 + X^2 + X + 1$:

$$\begin{aligned}
f(X) &= (X^2 + 1) + X(X^2 + 1) \\
(f(X) - (X^2 + 1))^2 &= X^2(X^2 + 1)^2 \\
(f(X) - p(X))^2 &= X^2(q(X))^2
\end{aligned}$$

□

Exercise R.4.5. Let $A = k[X, Y]/(Y^2 - X^2 - X^3)$. Prove that the normalization of A is $k[t]$ where $t = Y/X$.

Proof. $A = k[X, Y]/(Y^2 - X^2 - X^3)$, express X and Y in terms of t :

Since $t = Y/X$ then $Y = tX$, and combined with $Y^2 = X^2 + X^3$, then

$$\begin{aligned}
(tX)^2 &= X^2 + X^3 \\
t^2X^2 &= X^2 + X^3, \quad \text{assuming } X \neq 0 : \\
t^2 &= 1 + X, \quad \text{thus} \\
X &= t^2 - 1 \in k[X]
\end{aligned}$$

Then, $Y = tX = t(t^2 - 1) = t^3 - t \in k[X]$.

Hence $X, Y \in k[X]$.

Therefore, $k[X, Y]/(Y^2 - X^2 - X^3) \subseteq k[t]$.

By R.4.6 (Noether normalization lemma), to show that $k[t]$ is the *normalization*, must show that $k[t]$ is *integral* over A .

From $X = t^2 - 1 \implies t^2 - 1 - X = 0 \implies t^2 - (1 + X) = 0$.

$(1 + X) \in A$, so t satisfies the monic polynomial

$$P(T) = T^2 - (1 + X) \in A[T]$$

Thus t is integral over A .

Since $k[t]$ is generated by t over k , and $k \subset A$, then the entire ring $k[t]$ is integral over A .

Since $k[t]$ is a polynomial ring over a field, which is a UFD, it is integrally closed (since all UFD are integrally closed).

$\text{Frac} A = k(X, Y)$, since $X = t^2 - 1$, $Y = t^3 - t \implies k(X, Y) \subseteq k(t)$
and $t = Y/X \in k(X, Y)$, thus $k(X, Y) = k(t)$.

Since $k[t]$ is integrally closed and is the integral closure of A in its fraction field $k(t)$, we conclude that the normalization of A is $k[t]$. $\square$

Exercise R.4.9. k a field, $A = k[X, Y, Z]/(X^2 - Y^3 - 1, XZ - 1)$, find $\alpha, \beta \in k$ such that A is integral over $B = k[X + \alpha Y + \beta Z]$, and write a set of generators of A as a B -module.

Proof. (want to find a linear combination of the coordinates such that the original variables satisfy monic polynomials over the new ring B)

The relations defining A are

$$\begin{aligned} X^2 - Y^3 - 1 = 0 &\implies Y^3 = X^2 - 1 \quad (*) \\ XZ - 1 = 0 &\implies Z = 1/X = X^{-1} \end{aligned}$$

Thus A can be denoted as $A = k[X, Y, X^{-1}]/(Y^3 - X^2 - 1)$.

Now, Y is integral over $k[X]$, since $Y^3 - (X^2 - 1) = 0$ is monic in Y with coefficients in $k[X]$.

Z is not integral over $k[X]$, since $Z = 1/X$ and X is not a unit in $k[X]$.

Choose $\alpha, \beta \in k$ such that X (and thus Z) becomes integral over B :

set $\alpha = 0$, $\beta = 1 \implies B = k[X + \alpha Y + \beta Z] = k[X + Z]$.

Let $w = X + Z$; since $XZ = 1$, we have

$$w = X + \frac{1}{X} \implies Xw = X^2 + 1 \implies X^2 - wX + 1 = 0 \quad (**)$$

which is monic with coefficients in $k[w]$, thus X is integral over B .

Since $Z = w - X$, Z is also integral.

Generators of A as a B -module:

we have $B = k[w]$ with $w = X + Z$.

From $(**)$ we have $X^2 - wX + 1 = 0$, so $X^2 = wX - 1$.

Thus any polynomial in X can be reduced to a linear form $b_1X + b_0$ with $b_i \in k[w]$. Hence it's partial basis is $\{1, X\}$.

Fitting X^2 into $(*)$,

$$\begin{aligned} X^2 - Y^3 - 1 &= 0 \\ Y^3 &= X^2 - 1 \\ Y^3 &= wX - 2 \end{aligned}$$

thus any power of Y higher than 2 can be reduced (eg. $Y^4 = Y(wX - 2) = w(XY) - 2Y$).

So its partial basis is $\{1, Y, Y^2\}$.

For Z , since $XZ = 1$ and $w = X + Z \implies Z = w - X$, thus Z is a B -linear combination of $\{1, X\}$.

Combining the previous partial basis, the generators are

$$\{1, X\} \times \{1, Y, Y^2\} = \{1, Y, Y^2, X, XY, XY^2\}$$

□

8.5 Exercises Chapter 6

Exercise R.6.3.a. Let $A = A' \times A''$; prove that A' and A'' are rings of fractions of A .

Proof. ring of fractions of $A = S^{-1}A$, so want to prove that $S^{-1}A \cong A'$.

Localization map

$$\begin{aligned} \psi : A &\longrightarrow A' \\ (a', a'') &\longmapsto a' \end{aligned}$$

note that ψ is surjective, since $\forall a' \in A', \psi(a', 0) = a'$.

Let the multiplicative set S be $S = \{e_1\}$ with $e_1 = (1, 0)$.

Want $\underbrace{S^{-1}A}_{\text{localization}} \cong A'$.

In $S^{-1}A$, $x \in A$ maps to 0 iff $sx = 0$ for some $s \in S$.

Let $x = (a', a'') \in A$; then $sx = 0 \implies s \cdot (a', a'') = 0$,

with $s \in S$, so $s = (1, 0)$, hence

$$(1, 0) \cdot (a', a'') = (0, 0)$$

$$\implies (a', 0) = (0, 0)$$

which implies $a' = 0$, so that $(a', a'') = (0, a'')$.

$\implies$ the elements that become zero in the localization are of the form $(0, a'')$

$$\ker(\psi) = \{(a', a'') \in A \mid \psi(a', a'') = 0\} = \{(0, a'') \mid a'' \in A''\}$$

By the 1st isomorphism theorem,

$$\begin{array}{ccc} A & \xrightarrow{\psi} & A' \\ \phi \searrow & & \nearrow \eta \\ & \frac{A}{\ker(\psi)} & \end{array}$$

$$\implies \frac{A}{\ker(\psi)} \cong A'$$

Take the localization map

$$\begin{aligned}\phi : A &\longrightarrow S^{-1}A \\ a &\longmapsto a/1\end{aligned}$$

$$\ker(\phi) = \{x \in A \mid sx = 0 \text{ for some } s \in S\}$$

and we've seen that $a' = 0$, so $x \in A \Rightarrow x = (a', a'') = (0, a'')$

$$\ker(\phi) = \{(0, a'') \mid a'' \in A''\}$$

which is the same as $\ker(\psi)$; $\ker(\psi) = \ker(\phi)$.

$\Rightarrow A'$ and $S^{-1}A$ are both surjective images of A with exact same kernel, thus $A' \cong S^{-1}A$. $\square$

Exercise R.6.4. a. Give an example of a ring A and distinct multiplicative sets S, T such that $S^{-1}A = T^{-1}A$.

b. Prove that for fixed S , there is a maximal multiplicative set T with this property defined by

$$T = \{t \in A \mid at \in S \text{ for some } a \in A\}$$

Proof. a. (*saturated sets*)

General rule of saturation:

two multiplicative sets S, T yield the same localization, $S^{-1}A = T^{-1}A$, iff they have the same saturation.

Saturation of S : $\hat{S}$, set of all elements in A that divide some element of S :

$$\hat{S} = \{a \in A \mid \exists b \in A \text{ s.th. } ab \in S\}$$

Example 1: $A = \mathbb{Z}$,

$$S = \{2^n \mid n \geq 0\} = \{1, 2, 4, 8, 16, \dots\}$$

localization:

$$S^{-1}\mathbb{Z} = \left\{\frac{a}{2^n} \mid a \in \mathbb{Z}, n \in \mathbb{N}\right\}$$

$$T = \{\pm 1, \pm 2, \pm 4, \pm 8, \pm 16, \dots\}$$

Notice that despite $S \neq T$, we have $S^{-1}\mathbb{Z} = T^{-1}\mathbb{Z}$, since once $n \in \mathbb{Z}$ is invertible, $-n$ is automatically invertible: $(-n)^{-1} = -(n^{-1})$.

Example 2: $A = k[x]$,

$$S = \{x^n \mid n \geq 0\}$$

$$T = \{(2x)^n \mid n \geq 0\}$$

$$\implies S^{-1}A = T^{-1}A = k[x, x^{-1}]$$

b. Show that T is a multiplicative set:

(must contain 1 and be closed under multiplication)

Since $1 \in A$ and $1 \cdot 1 = 1 \in S$, then $1 \in T$.

Let $t_1, t_2 \in T$; by definition of T ,

$$\exists a_1, a_2 \in A \text{ such that } a_1 t_1, a_2 t_2 \in S$$

Since S a multiplicative set, $(a_1 t_1)(a_2 t_2) \in S$

rearrange it: $(a_1 a_2)(t_1 t_2) \in S$.

Since $a_1 a_2 \in A$, $t_1 t_2 \in T$.

Thus T is a multiplicative set.

Next, we show that T is maximal:

Suppose U is the maximal instead of T , such that $U^{-1}A \cong S^{-1}A$.

Let $u \in U$; the image of u in $S^{-1}A$ must be a unit.

Units in $S^{-1}A$ are the fractions $\frac{s}{a}$ such that their inverse exists.

$$\exists x = \frac{a}{s} \text{ such that } u \frac{a}{s} = \frac{1}{1}$$

$$\implies \frac{ua}{s} = \frac{1}{1} \implies \exists s' \in S \text{ such that } s'(ua - s) = 0$$

$$\implies s'ua = s's \in S$$

since $s, s' \in S$, their product is also in S .

Let $b = s'a$, then $ub \in S$.

By definition of T , $u \in T$.

Therefore, $U \subseteq T$; thus T is maximal.

□

References

- [1] M. F. Atiyah and I. G. MacDonald. Introduction to Commutative Algebra, 1969.
- [2] Miles Reid. Undergraduate Commutative Algebra, 1995.
- [3] Steven Kleiman. Commutative Algebra - MIT Open-CourseWare, 2008. <https://ocw.mit.edu/courses/18-705-commutative-algebra-fall-2008/>.